



Research paper

Kinematic and dynamic modeling of cable-object interference and wrapping in complex geometrical-shaped cable-driven parallel robots

Dipankar Bhattacharya^{ID*}, T.K. Cheung, Yaxiang Wang^{ID}, Darwin Lau^{ID}

Department of Mechanical and Automation Engineering, The Chinese University of Hong Kong, Ma Liu Shui, Shatin, Hong Kong



ARTICLE INFO

Keywords:

Cable-driven parallel robots
Non-uniform rational B-splines
Cable wrapping
Geodesics
Interference handling
Kinematics
Dynamics
Friction modeling

ABSTRACT

Cable-Driven Parallel Robots (CDPRs) use cables as actuators to maneuver rigid mobile-platform in a parallel mechanism setup. Typically, CDPR kinematic and dynamic models avoid cable-object (cable-mobile-platform and cable-obstacle) interferences to prevent sudden cable tension changes that could deviate the end-effector's trajectory. However, allowing these interferences can lead to cable wrapping, where cables wrap around complex-shaped surfaces upon contact, enhancing the CDPR's workspace and reducing its footprint. Despite the potential benefits, there currently exists no kinematic and dynamic model that effectively incorporates cable wrapping around such complex-shaped surfaces. This paper introduces a novel numerical-based kinematic and dynamic modeling framework for CDPRs that detects and then manages cable wrapping around mobile-platform and multiple obstacles with the assumption that the cables remain taut and for every position along the cable, there is a unique and smooth way to describe its location on the surface. Simulation and hardware results on various complex-shaped mobile-platform and obstacles show that the proposed model framework can be conveniently and effectively applied to the real-time modeling of cable wrapping. Code and videos available at: <https://github.com/bhattner143/GeoWrapSim-CDPR.git>.

1. Introduction

Cable-Driven Parallel Robots (CDPRs) are a type of parallel manipulator where cables actuate the movement of rigid body mobile-platform. These robots offer benefits such as low inertia, large potential workspaces, high reconfigurability, and transportability. Consequently, they are widely studied and applied across various fields, including manufacturing [1,2], building construction [3,4], rapid prototyping [5], and rehabilitation [6,7]. Examples such as the Skycam [8] and the RoboCrane [1], which is used to remove paint from military aircraft, BMArm (Fig. 1(a) and (b)) [9] for window cleaning and painting, and SpiderX52 (Fig. 1(c)), a 4-Degrees of Freedom (DoF) RM-X52 open manipulator on the spatial cable-driven robot end-effector [9], for industrial object manipulation tasks, demonstrate the utility of CDPRs in having large workspaces suitable for commercial applications.

CDPR modeling is typically divided into two main categories: kinematics and dynamics, where each serves a distinct purpose in understanding and controlling the behavior of CDPRs. *CDPR kinematics* involves the geometry and motion of the robot, while *CDPR dynamics* examines the forces, torques, and accelerations affecting the system. *CDPR kinematics* includes *Inverse Kinematics (IK)*, determining cable lengths for specific poses, and *Forward Kinematics (FK)*, estimating poses from cable lengths. *CDPR dynamics*

* Corresponding author.

E-mail addresses: dipankarbhattacharya@cuhk.edu.hk (D. Bhattacharya), kitchung1480@link.cuhk.edu.hk (T.K. Cheung), yaxiangw@link.cuhk.edu.hk (Y. Wang), darwinlau@cuhk.edu.hk (D. Lau).

<https://doi.org/10.1016/j.mechmachtheory.2025.106092>

Received 14 February 2025; Received in revised form 25 April 2025; Accepted 24 May 2025

Available online 17 July 2025

0094-114X/© 2025 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

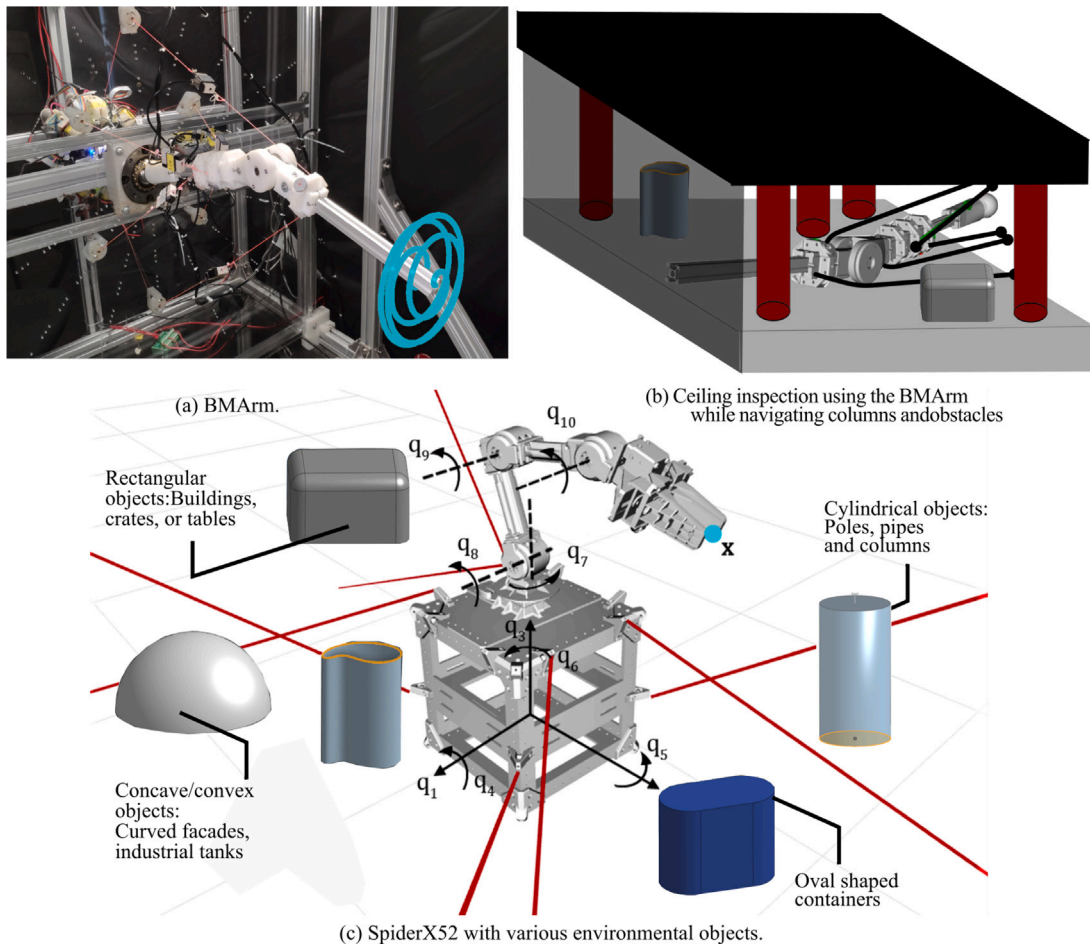


Fig. 1. Examples of Cable-Driven Parallel Robots (CDPRs) performing various practical applications. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

is divided into *Inverse Dynamics (ID)*, which determines the cable forces needed for desired end-effector motions, and *Forward Dynamics (FD)*, which predicts CDPR motions from known cable forces and tensions.

The kinematic and dynamic models of CDPRs are often simplified by neglecting phenomena such as cable interferences, which occur when cables come into contact with each other or with nearby objects or mobile-platform. The cable interference with nearby objects or mobile-platform can be classified into two categories: *cable-mobile-platform interference* and *cable-obstacle interference* [10], collectively referred to as *object-interference*. During CDPR operation, if object-interference occurs within the workspace, such as BMArm (Fig. 1(a)) cable hitting its mobile-platform, vertical columns or obstacles while doing ceiling inspection (Fig. 1(b)) or SpiderX52's cable colliding with the object(s) (Fig. 1(c)), then it can cause an instantaneous change in cable tension, leading to a deviation in the end-effector trajectory of the entire CDPR. This deviation can severely impact the performance of the CDPR. Additionally, such interferences make path planning in cluttered environments much more challenging. Therefore, detecting object-interferences is an essential part of CDPR modeling.

However, detecting object-interference in CDPRs presents significant challenges, particularly when dealing with complex-shaped objects such as obstacles or mobile-platform within the workspace. The intricate geometries of these objects can create unpredictable interaction points where cables might contact or entangle with the environment. This scenario complicates the modeling and simulation processes needed to accurately predict potential interferences. Nevertheless, implementing *interference detection algorithms* dedicated to avoiding interferences within a CDPR system would inevitably lead to a reduction in the usable workspace of the CDPR.

Hence, instead of strictly avoiding them, a more sophisticated approach could involve detecting interferences and allowing them within the CDPR model. Permitting interactions between a cable and nearby complex-shaped objects, such as obstacles and mobile-platform, leads to a phenomenon known as *cable wrapping* [11]. This occurs when a cable begins to wrap around an object's surface upon contact. Allowing interferences in the form of cable wrapping on CDPR objects can provide multiple advantages: (1) It enables the generation of larger moments, which expands the rotational workspace [12,13]. (2) Introducing wrapping into the CDPR model can lead to a more compact robot design by reducing its overall footprint. (3) Wrapping allows the CDPR to maneuver

more effectively within confined spaces. For example, in Fig. 1(b), the BMArm, operating within a confined space, can effectively navigate columns (red vertical structures) and other obstacles to inspect the ceiling when wrapping is permitted. Similarly, in Fig. 1(c) allowing the SpiderX52's cables to wrap around the objects within its workspace can showcase how wrapping can enhance the robot's ability to operate in cluttered environments. Despite the numerous advantages associated with cable wrapping, the practice of permitting interferences to allow for wrapping in CDPRs has been largely neglected.

In the field of interference detection algorithms, pioneering work by Merlet et al. and Blanchet et al. has introduced interval analysis-based algorithms to detect collisions between cylindrical-shaped bodies and cables in CDPRs [14,15]. Blanchet et al. further extended these algorithms to address various kinds of obstacle- and self-interferences, which limit the motion of CDPRs [15]. Assuming fixed orientations, Perreault et al. developed algorithms that efficiently detect and avoid cable-cable and cable-mobile-platform interferences [16]. However, these algorithms do not address the complexities arising from interactions with multiple objects or objects of more complex geometries. Rushton et al. introduced an online algorithm that utilizes a *cable state tree* structure to continuously update the status of all cables prone to colliding with various circular objects due to end-effector motion [17]. While effective, this algorithm is limited to scenarios involving simple circular objects, which can be easily defined geometrically. In summary, while existing works have made significant contributions to the detection and avoidance of cable interferences in CDPRs, they have not fully addressed the challenges associated with detecting interferences involving multiple objects or mobile-platform with complex geometries.

Likewise, current research on modeling cable wrapping in CDPR kinematics and dynamics, has been rather limited and typically focuses on simple geometric surfaces. This modeling often utilizes the concept of *geodesics*, defined as the shortest paths between two points on a surface, equivalent to a straight line on a plane [18,19]. However, the application of geodesics in CDPR modeling remains underexplored. Several studies have addressed cable wrapping in CDPRs with assumptions of taut wrapping. Lei et al. developed an analytical Cable Wrapping Geodesic (CWG) model for a single cylindrical mobile-platform CDPR [20,21]. By assuming frictionless surfaces, Xiong et al. solved the statics and kinematics of cables wrapped around simple surfaces [22,23]. While their method conceptually extends to any frictionless, complex surface, it is limited to cylinders and spheres, lacking a generalized methodology or formulations for more complex geometries. Moreover, deriving equations for each individual surface is impractical and labor-intensive. Zou et al. developed a data-driven kinematic control scheme, trained using experimental cable wrapping data from simple surfaces like cylinders and spheres, limiting its use to these surfaces [24]. Additionally, the work does not address the challenge of collecting data experimentally from complex surfaces.

Sun et al. explored cable wrapping to enhance the kinematics and workspace of CDPRs with a circular end-effector [13]. Despite these advancements, the methodologies are not generalizable to multiple complex geometric surfaces with friction, and they lack interference detection. They assume continuous cable wrapping, which is impractical due to the potential for cable interference and detachment during CDPR motion.

In the field of multiple object wrapping, Rushton et al. developed a planar variable-structure CDPR for wrapping and collision with multiple circular objects, though limited to circular shapes and planar motion [17]. Lloyd et al. proposed a method for computing wrapping paths across arbitrary surfaces, such as those defined by bone geometry [25], and Scholz et al. introduced the Natural Geodesic Variation (NGV) method for efficiently computing musculo-tendon paths on curved surfaces like torus and polynomial patches [26]. However both methods lack interference detection algorithm. Additionally, Lloyd et al.'s method only demonstrates wrapping on single surfaces, and both approaches are limited to frictionless surfaces. Muñoz-Pepi et al. used nonlinear optimization for muscle wrapping simulations over two cylinders, allowing state transitions but lacking a dedicated detection algorithm and being computationally intensive for complex shapes [27]. In summary, previous studies on interference detection and wrapping are overly simplistic and not well-suited for multiple complex-shaped geometrical objects.

To contribute to the research on CDPRs interacting with multiple complex-shaped objects in presence of friction, this paper aims is to design a novel kinematic and dynamic framework that detects interference between CDPR cables and multiple complex geometrical-shaped objects, and then allow the cables to wrap around these objects. Three objectives are set to achieve the aim: First, to numerically obtain a CWG model that can accurately represent the 3D motion of CDPR cables interacting with complex-shaped objects. Second, to implement an algorithm that detects cable interferences and detachments with multiple objects concurrently. Third, to apply the CWG model to effectively address the Cable Wrapping Geodesic-based Inverse Kinematics (CWGIK) and friction-compensated Cable Wrapping Geodesic-based Inverse Dynamics (CWGID) problems in complex-shaped CDPRs.

To achieve the outlined goals, the framework design involved several steps: First, complex-shaped CDPRs are modeled using Non-uniform rational B-splines (NURBS), ensuring geometric continuity vital for dynamic simulations and representing a wide range of shapes with a single mathematical description. Second, CWG differential equations based on the minimum cable length criteria are obtained for the complex-shaped CDPRs to understand cable interaction relative to the object geometries. Third, a CWG model is developed by numerically solving these equations through optimization, crucial for accurately simulating real-world cable motion around complex shapes. Fourth, a cable multi-object interference detection algorithm is implemented with the CWG model to monitor and manage cable interactions, ensuring prompt detection of interferences and detachments. Fifth, this algorithm and the CWG model are applied to solve the CWGIK and friction-compensated CWGID problems, enhancing the operational capabilities of the CDPRs with complex-shaped objects. Sixth, the capability of the framework is illustrated by simulations on BMWrapArm [28], a 3-DOF spherical joint CDPR, with various complex-shaped mobile-platform and obstacles, through Cable-Robot Analysis and Simulation Platform for Research (CASPR) software. Finally, experimental validations on the BMWrapArm are also conducted to show the ability of the framework to operate on hardware. The proposed method is developed to facilitate the understanding and analysis of CDPRs involving general cable wrapping scenarios.

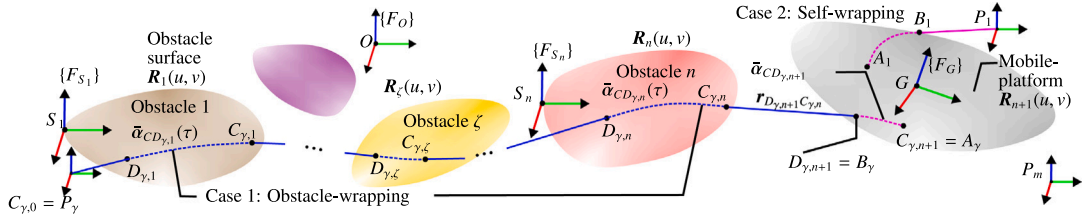


Fig. 2. System description of a CDPR with arbitrary obstacle and mobile-platform bodies.

The remainder of this article is organized as follows. Section 2 offers a system description of a single mobile-platform CDPR with cable wrapping on multiple complex-geometrically shaped mobile-platform and obstacles. Section 3 introduces CWG model for modeling cable wrapping. Section 4 presents a novel algorithm for detecting cable-object interference with multiple complex geometrical-shaped objects. Leveraging the concepts from Sections 2 to 4, Sections 5 and 6 introduce the CWGIK and CWGID models for multiple complex-shaped obstacles and mobile-platform, respectively, and then introduces cable surface friction modeling to calculate the actual motor side ID force, compensating friction. Sections 7 and 8 discuss simulation and experimental results, respectively. Finally, Section 9 summarizes the conclusions and outlines future work.

2. Description of complex geometrical-shaped CDPR

Consider a CDPR as shown in Fig. 2, which features a complex geometrically shaped mobile-platform operating amidst total k different complex-shaped obstacles (Fig. 2), lying within its workspace, located at S_ζ . At any given time step, cable γ of the CDPR is interfering with $n \leq k$ obstacles. These objects (obstacles and mobile-platform) can range from complex NURBS to simpler surfaces such as tori and cylinders. The surface geometries of such objects are defined by the following parametric equations

$$\mathbf{R}_\zeta(u, v) = [x_{o_\zeta}(u, v), y_{o_\zeta}(u, v), z_{o_\zeta}(u, v)]^T \in \mathbb{R}^3 \quad (1)$$

where

- subscript ζ refer to both the obstacles and the mobile-platform, such that $\zeta = 1, \dots, n+1$, where the first n indices represent the obstacles, $(n+1)$ index refers to the mobile-platform, and n is pose $q(t)$ dependent,
- (u, v) are parameters within a domain $\mathcal{D}_\zeta$, given by $\mathcal{D}_\zeta = \{(u, v) \mid a \leq u \leq b, c \leq v \leq d, (a, b, c, d) \in \mathbb{R}\}$, and
- $\mathbf{R}_\zeta(u, v) : (u, v) \rightarrow (x, y, z)$ maps the uv -plane in $\mathbb{R}^2$ into $\mathbb{R}^3$, represents non-singular, differentiable vector-valued functions of (u, v) such that $\mathbf{R}_\zeta(u, v)$ is at least twice continuously differentiable over $\mathcal{D}_\zeta$ i.e., $\mathbf{R}_\zeta(u, v) \in \mathbb{C}^2(\mathcal{D}_\zeta)$.

Within the workspace, the CDPR mobile-platform can be moved by retracting or releasing m number of massless cables (Fig. 2), numbered by $\gamma = 1, \dots, m$, through winches actuated by motors on the fixed platform. The pose of the mobile-platform at any given instant of time t can be defined by the generalized coordinates $q(t) \in \mathbb{R}^{n_d}$, where n_d denotes the number of DoF of the mobile-platform. The change in pose q affects multiple complex shaped object-interferences and detachments concurrently. If interference with objects indexed by ζ is detected, then the cables wrap around both the obstacle and the mobile-platform. Here, $\zeta = 1$ is assigned to the mobile-platform, and $\zeta = 2, \dots, k+1$ correspond to the obstacle indices. This phenomenon includes *multi-wrapping*, *obstacle-wrapping*, and *self-wrapping*, collectively referred to as *object-wrapping*. The *object-wrapping state* η_ζ associated with these $k+1$ complex-shaped objects can be written as

$$\eta_\zeta = \begin{cases} 1 & \text{if wrapping at object } \zeta, \\ 0 & \text{if no wrapping at object } \zeta, \end{cases} \quad \zeta = 1, \dots, k+1. \quad (2)$$

At any given pose q , if exactly n obstacles and one mobile-platform are wrapped, then

$$n+1 = \sum_{i=1}^{k+1} \eta_i. \quad (3)$$

For example, with four objects ($k = 3$ obstacles and one mobile-platform), if the cable wraps around the mobile-platform ($\zeta = 1$) and the third obstacle ($\zeta = 4$), then $\eta_1 = 1, \eta_2 = 0, \eta_3 = 0, \eta_4 = 1$. Thus, two objects are wrapped ($n+1 = 2$).

Assumptions. In this work, after obtaining η_ζ , the following assumptions are made to determine the cable-object wrapping

1. Location of the objects $\mathbf{R}_\zeta(u, v)$, given by S_ζ are known.
2. $\mathbf{R}_\zeta(u, v)$ is at least twice continuously differentiable over $\mathcal{D}_\zeta$ i.e., $\mathbf{R}_\zeta(u, v) \in \mathbb{C}^2(\mathcal{D}_\zeta)$.
3. Viewed from the mobile-platform surface, a cable wrapping curve $\alpha_{CD_{\gamma,\zeta}}(\tau)$, parameterized by τ , lying on the object $\mathbf{R}_\zeta(u, v)$ between the cable entry $C_{\gamma,\zeta}$ and exit $D_{\gamma,\zeta}$, satisfies the condition that for each point $\tau \in [\tau_{C_{\gamma,\zeta}}, \tau_{D_{\gamma,\zeta}}]$, there exists unique, smooth pair of parameters $(u(\tau), v(\tau)) \in \mathcal{D}_\zeta$ such that $\alpha_{CD_{\gamma,\zeta}}(\tau) = \mathbf{R}_\zeta(u(\tau), v(\tau))$, where $\alpha_{CD_{\gamma,\zeta}}(\tau)$ remains continuously differentiable, ensuring no gaps or discontinuities.

4. The cable force f_γ is positive and constant along its length, ensuring there is no slack: $f_\gamma(\tau) > 0, \forall \tau \in [\tau_{C_{\gamma,\zeta}}, \tau_{D_{\gamma,\zeta}}]$.

Remark 1. At any given pose q , if the wrapped cable $\alpha_{CD_{\gamma,\zeta}}(\tau)$ path satisfies Assumptions 3–4, then it is known as CWG $\bar{\alpha}_{CD_{\gamma,\zeta}}$, which is the shortest path on $R_\zeta(u, v)$ between $C_{\gamma,\zeta}$ and $D_{\gamma,\zeta}$ among all other wrapping configurations [18,19] i.e.

$$\bar{\alpha}_{CD_{\gamma,\zeta}}(\tau) = \underset{\alpha}{\operatorname{argmin}} \int_{\tau_{C_{\gamma,\zeta}}}^{\tau_{D_{\gamma,\zeta}}} \left\| \frac{d\alpha(\tau)}{d\tau} \right\| d\tau, \quad (4)$$

subject to boundary condition $\bar{\alpha}_{CD_{\gamma,\zeta}}(\tau_{C_{\gamma,\zeta}}) = r_{SC_\zeta}$ and $\bar{\alpha}_{CD_{\gamma,\zeta}}(\tau_{D_{\gamma,\zeta}}) = r_{SD_\zeta}$.

Fig. 2 illustrates the object wrapping, such that the wrapped part of the cables is determined by the CWG model (as presented in Section 3). To determine the CWG for object-wrapping, the following frames are defined

- $\{F_O\}$ and $\{F_G\}$ represent the inertial frame and the body frame fixed at the mobile-platform's CoM located at G , respectively.
- $\{F_{S_\zeta}\}$ refers to the obstacle frame located at the origin of obstacle ζ .

3. Cable wrapping geodesic model

This section discusses the methodology for obtaining the CWG model in the case of object-wrapping. Here, one end of the cable is fixed to the base and the other end is connected to the mobile CDPR mobile-platform. As the CDPR operates, the cables wrap both the static obstacles and the moving mobile-platform. The model is used to determine the length of all the cables that wrap around the object surfaces. To find the cable length of cable γ , the object-wrapping profile $(A_\gamma D_{\gamma,n+1} C_{\gamma,n} D_{\gamma,n} \cdots C_{\gamma,\zeta} D_{\gamma,\zeta} \cdots C_{\gamma,1} D_{\gamma,1} P_\gamma)$ can be written as (Fig. 2) a combination of

- parameterized CWGs on the mobile-platform and the obstacles denoted by $\bar{\alpha}_{CD_{\gamma,\zeta}}(\tau)$, such that $\zeta = 1, \dots, n+1$ and $C_{\gamma,n+1} = A_\gamma$, and
- straight cable segments described by the direction vectors from $D_{\gamma,\zeta+1}$ to $C_{\gamma,\zeta}$ and $D_{\gamma,\zeta}$ to $C_{\gamma,\zeta-1}$ denoted as $r_{D_{\gamma,\zeta+1}C_{\gamma,\zeta}}$ and $r_{D_{\gamma,\zeta}C_{\gamma,\zeta-1}}$, respectively where $\zeta = 1, \dots, n$ and $C_{\gamma,0} = P_\gamma$.

Remark 2. During the operation of a CDPR, under Assumptions 1–4, the CWGs on each object (obstacles and mobile-platform) must remain tangential to the vectors connecting the cable entry and exit points. For a taut and smooth cable, these tangential conditions ensure that the cable maintains a continuous gradient while wrapping around the objects. Specifically, the vector $r_{D_{\gamma,\zeta+1}C_{\gamma,\zeta}}$ must be tangent to the cable curve at the entry point $C_{\gamma,\zeta}$, and the vector $r_{D_{\gamma,\zeta}C_{\gamma,\zeta-1}}$ must be tangent to the cable curve at the exit point $D_{\gamma,\zeta}$ for all objects indexed by $\zeta = 1, \dots, n+1$. The first n tangential conditions correspond to obstacle-wrapping, ensuring smooth transitions around external objects, while the $n+1$ th condition specifically addresses self-wrapping around the robot's mobile-platform. These tangential conditions must hold for all joint configurations q to ensure proper cable routing and smooth, continuous operation of the system.

To facilitate easier reference and reduce complexity in the expressions, we introduce the following notational simplifications for the analysis:

- Subscript ζ is dropped since the below analysis is done on object ζ , and it is applicable for all objects indexed from $1, \dots, n+1$.
- All vectors will be assumed to be with respect to the frame $\{F_{S_\zeta}\}$ unless otherwise indicated by a left-hand side superscript. This assumption is based on the fact that object ζ is the primary subject of the analysis.
- Subscript γ is also removed, as the subsequent analysis is applicable for all cables involved in the wrapping process.

3.1. Defining complex geometrical-shaped objects

During CDPR motion, the parametric equations $R(u, v)$ of complex-shaped object surface can be precisely defined by NURBS surface, which is a mathematical model used to represent 3D geometry [29]. A NURBS surface parametric equations can be defined as

$$R(u, v) = \frac{\sum_{i=0}^{n_u} \sum_{j=0}^{n_v} N_{i,d_u}(u) N_{j,d_v}(v) w_{ij} \phi_{ij}}{\sum_{i=0}^{n_u} \sum_{j=0}^{n_v} N_{i,d_u}(u) N_{j,d_v}(v) w_{ij}}, \quad (5)$$

where

- $\phi_{ij} \in \mathbb{R}^3$, $i = 0, \dots, n_u$, and $j = 0, \dots, n_v$ are the control points that influence the shape of the NURBS, usually do not lie on the surface itself but act like magnets pulling the surface into shape.
- $n_u + 1$ and $n_v + 1$ are the numbers of control points in the u and v directions, respectively.
- p_u and p_v are the degrees of the surface in the u and v directions, respectively. They can be used to define the level of curvature such that a higher degree allows for a smoother and more complex surface.

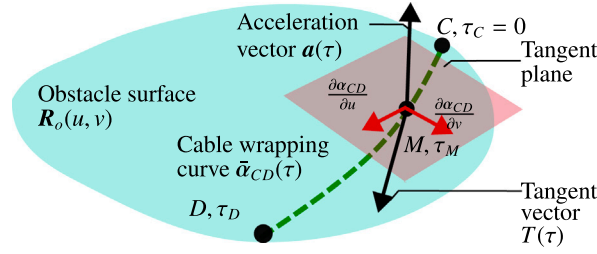


Fig. 3. Description of the CWG tangent (or velocity) and acceleration vectors.

- $N_{i,p_u}(u)$ and $N_{j,p_v}(v)$ are the Bezier-spline (B-Spline) basis functions for the u and v parameters, respectively, which can be determined by *Cox-de Boor recursion formula* [30]. These basis functions play a critical role in forming the NURBS surface, as they define how much influence each control point has over specific regions of the surface based on the parameters u and v . Basis function $N_{i,p_u}(u)$ can be defined as follows

$$N_{i,k}(u) = \begin{cases} 1 & \text{if } u_i \leq u < u_{i+1}, \quad k = 0, \\ 0 & \text{otherwise, } k = 0, \\ \frac{u - u_i}{u_{i+k} - u_i} N_{i,k-1}(u) + \frac{u_{i+k+1} - u}{u_{i+k+1} - u_{i+1}} N_{i+1,k-1}(u) & \text{if } 1 \leq k \leq p_u, \end{cases} \quad (6)$$

where u_i are the knot values in the knot vector $\mathbf{U}$. At $k = 0$, the basis function $N_{i,0}(u)$ is a step function, which takes the value 1 if $u_i \leq u < u_{i+1}$ and 0 otherwise. For $k > 0$, the recursion blends two adjacent lower-degree basis functions, $N_{i,k-1}(u)$ and $N_{i+1,k-1}(u)$, using linear interpolation. The term $\frac{u - u_i}{u_{i+k} - u_i}$ in the recursion is the weight for $N_{i,k-1}(u)$, representing the relative position of u within the interval $[u_i, u_{i+k}]$. When $u = u_i$, this term is zero, and as u approaches u_{i+k} , it increases linearly to 1. The term $\frac{u_{i+k+1} - u}{u_{i+k+1} - u_{i+1}}$ is the weight for $N_{i+1,k-1}(u)$, representing how far u is from u_{i+1} within the interval $[u_{i+1}, u_{i+k+1}]$. When $u = u_{i+k+1}$, this term becomes zero, and it linearly decreases from 1 as u moves away from u_{i+1} . These weights ensure that the basis functions are smoothly blended over the intervals, with nonzero values only within the range $[u_i, u_{i+k+1}]$. This blending ensures local support and continuity of the basis functions.

- The knot vectors $\mathbf{U} = [u_0, u_1, \dots, u_{n_u+p_u+1}]$ and $\mathbf{V} = [v_0, v_1, \dots, v_{n_v+p_v+1}]$ are non-decreasing sequences of parameter values ($u_{i+1} \geq u_i \forall i$ and $v_{j+1} \geq v_j \forall j$) that control how the basis functions and control points influence the NURBS surface. Uniform knot vectors result in equally spaced surface segments, while non-uniform knot vectors allow for irregular spacing, offering greater flexibility in shaping the surface by allowing localized control. Each unique knot value can have a *multiplicity*, with higher multiplicity reducing surface continuity at that parameter. Knot vectors govern the parametric spacing, continuity, and control point influence in the u - and v -directions.
- w_{ij} are the weights for the control points $\phi_{ij} \in \mathbb{R}^3$, which can be adjusted to fine-tune influence of each control point on the shape of the surface.

The compact form of (5) can be written as

$$\mathbf{R}(u, v) = [\mathbf{N}_u(\mathbf{W}\phi_x)\mathbf{N}_v^T, \mathbf{N}_u(\mathbf{W}\phi_y)\mathbf{N}_v^T, \mathbf{N}_u(\mathbf{W}\phi_z)\mathbf{N}_v^T]^T \in \mathbb{R}^3, \quad (7)$$

where

$$\begin{aligned} \bullet \mathbf{N}_u &= [N_{0,p_u}, N_{1,p_u}, \dots, N_{n_u,p_u}]^T \in \mathbb{R}^{(n_u+1)}, \quad \mathbf{N}_v = [N_{0,p_v}, N_{1,p_v}, \dots, N_{n_v,p_v}] \in \mathbb{R}^{(n_v+1)} \\ \bullet \mathbf{W} &= \begin{bmatrix} w_{0,0} & w_{0,1} & \dots & w_{0,n_v} \\ w_{1,0} & w_{1,1} & \dots & w_{1,n_v} \\ \vdots & \vdots & \ddots & \vdots \\ w_{n_u,0} & w_{n_u,1} & \dots & w_{n_u,n_v} \end{bmatrix} \in \mathbb{R}^{(n_u+1) \times (n_v+1)} \\ \bullet \phi_x, \phi_y, \text{ and } \phi_z &\in \mathbb{R}^{(n_u+1) \times (n_v+1)} \text{ are control point matrices storing } x, y, \text{ and } z \text{ components of the control point } \phi_{i,j}. \end{aligned}$$

Next, cable wrapping geodesic around $\mathbf{R}(u, v)$ is discussed.

3.2. Determining cable wrapping geodesic

Under Assumptions 1–4, by applying the Euler–Lagrange equations on the minimum length criteria of the cable wrapping, CWG partial differential equations can be obtained, and solved to determine the CWG parametric equations $\alpha_{CD}(\tau) \in \mathbf{R}(u, v)$ for $\tau \in [\tau_C, \tau_D]$. Hence, this section discusses a general methodology to obtain the CWG length and the differential equations.

Let M , given by $\mathbf{R} : (u(\tau_M), v(\tau_M)) \rightarrow (x(\tau_M), y(\tau_M), z(\tau_M))$, be a point on the object surface (Fig. 3), lying on the cable wrapping curve α_{CD} , such that $\tau_M \in [0, \tau_D]$, where, without loss of generality, τ_C , parameterizing the initial cable arc length, is assumed to be zero. The tangent space to the surface at point M is a plane spanned by orthogonal basis vectors

$$\mathbf{R}_u = \frac{\partial \alpha_{CD}}{\partial u}, \quad \mathbf{R}_v = \frac{\partial \alpha_{CD}}{\partial v}, \quad (8)$$

where the derivatives are evaluated at $(u(\tau_M), v(\tau_M))$. For any point on α_{CD} , the infinitesimal displacement vector $d\alpha_{CD}$ is given by

$$d\alpha_{CD} = \mathbf{R}_u du + \mathbf{R}_v dv, \quad (9)$$

where du and dv are coefficients of the orthogonal basis vectors. Hence, from (9), the infinitesimal arc length can be written as

$$ds = \|d\alpha_{CD}\|_2 = \sqrt{Edu^2 + Fdudv + Gdv^2}, \quad (10)$$

where $E = \mathbf{R}_u \cdot \mathbf{R}_u$, $G = \mathbf{R}_v \cdot \mathbf{R}_v$, $F = \mathbf{R}_u \cdot \mathbf{R}_v$, and $F = 0$, if $\mathbf{R}_u$ and $\mathbf{R}_v$ are orthonormal basis vectors. In (10), the expression under the root is known as the *first fundamental form*, which enables measurement of lengths on a surface.

Proposition 1. Given $\mathbf{R}_u$ and $\mathbf{R}_v$ of the complex-shaped object surface $\mathbf{R}(u, v)$, the cable wrapping curve length l_w of $\alpha_{CD}(\tau)$ on $\mathbf{R}(u, v)$ from $\tau_C = 0$ to τ_D can be defined by

$$l_w = \int_0^{s_D} ds = \int_0^{\tau_D} \frac{ds}{d\tau} d\tau = \int_0^{\tau_D} H(u(\tau), v(\tau), \bar{u}'(\tau), \bar{v}'(\tau)) d\tau, \quad (11)$$

where s_D is the arc-length at D , and

$$H = \sqrt{E\bar{u}'^2 + F\bar{u}'\bar{v}' + G\bar{v}'^2}, \quad (12)$$

with $\bar{u}' = \frac{d\bar{u}}{d\tau}$ and $\bar{v}' = \frac{d\bar{v}}{d\tau}$ represent the derivatives of the surface parameters with respect to τ . $\circ$

Proposition 2. If a path given by $\{u(\tau), v(\tau) | \tau \in [0, \tau_D]\}$ on the uv -plane minimizes the cable wrapping curve length functional $l_w(u, v)$, given by (11) then such a path is known as the optimal parameterized path, and it is given by $\{\bar{u}(\tau), \bar{v}(\tau) | \tau \in [0, \tau_D]\}$. Hence, the optimal cable length $\bar{l}_w$ can be written as

$$\bar{l}_w := l_w(\bar{u}, \bar{v}) = \min_{u(\tau), v(\tau)} \int_0^{\tau_D} H(u, v, u', v') d\tau, \quad (13)$$

where $\bar{l}_w$ is the length of CWG $\bar{\alpha}_{CD}(\tau)$, whose parametric equations can be obtained by the following

$$\bar{\alpha}_{CD}(\tau) = \mathbf{R}(\bar{u}(\tau), \bar{v}(\tau)) \quad \circ \quad (14)$$

To satisfy (13) or in other words to determine the CWG parametric equations (14), the optimal parameterized path $\{\bar{u}(\tau), \bar{v}(\tau) | \tau \in [0, \tau_D]\}$ on the uv -plane needs to be determined for the given surface $\mathbf{R}(u, v)$.

Proposition 3. An optimal parameterized path $\{\bar{u}(\tau), \bar{v}(\tau) | \tau \in [0, \tau_D]\}$ on the uv -plane that minimizes (11) to get $\bar{l}_w$ can be obtained by solving the following set of Euler–Lagrange equations

$$\frac{\partial H}{\partial \bar{u}} - \frac{d}{d\tau} \left(\frac{\partial H}{\partial \bar{u}'} \right) = 0, \quad \frac{\partial H}{\partial \bar{v}} - \frac{d}{d\tau} \left(\frac{\partial H}{\partial \bar{v}'} \right) = 0. \quad \circ \quad (15)$$

The Euler–Lagrange equations in (15), essential for determining the geodesic path, are challenging to solve analytically and numerically. To simplify and derive the *geodesic equations*, derivatives of (12) with respect to $\bar{u}$, $\bar{v}$, $\bar{u}'$, and $\bar{v}'$ can be written as

$$\begin{aligned} \frac{\partial H}{\partial \bar{u}} &= \frac{1}{2H} (E_{\bar{u}}(\bar{u}')^2 + 2F_{\bar{u}}\bar{u}'\bar{v}' + G_{\bar{u}}(\bar{v}')^2), & \frac{\partial H}{\partial \bar{u}'} &= \frac{1}{H} (E\bar{u}' + F\bar{v}'), \\ \frac{\partial H}{\partial \bar{v}} &= \frac{1}{2H} (E_{\bar{v}}(\bar{u}')^2 + 2F_{\bar{v}}\bar{u}'\bar{v}' + G_{\bar{v}}(\bar{v}')^2), & \frac{\partial H}{\partial \bar{v}'} &= \frac{1}{H} (F\bar{u}' + G\bar{v}'), \end{aligned} \quad (16)$$

where

$$E_{\bar{u}} = \frac{\partial E}{\partial \bar{u}}, \quad F_{\bar{u}} = \frac{\partial F}{\partial \bar{u}}, \quad G_{\bar{u}} = \frac{\partial G}{\partial \bar{u}}, \quad E_{\bar{v}} = \frac{\partial E}{\partial \bar{v}}, \quad F_{\bar{v}} = \frac{\partial F}{\partial \bar{v}}, \quad \text{and} \quad G_{\bar{v}} = \frac{\partial G}{\partial \bar{v}}. \quad (17)$$

Next, taking the derivatives of $\frac{\partial H}{\partial \bar{u}'}$ and $\frac{\partial H}{\partial \bar{v}'}$ in (17) with respect to τ and applying the product rule yields

$$\frac{d}{d\tau} \left(\frac{\partial H}{\partial \bar{u}'} \right) = \frac{(E\bar{u}' + F\bar{v}')'}{H} - \frac{(E\bar{u}' + F\bar{v}')H'}{H^2}, \quad \frac{d}{d\tau} \left(\frac{\partial H}{\partial \bar{v}'} \right) = \frac{(F\bar{u}' + G\bar{v}')'}{H} - \frac{(F\bar{u}' + G\bar{v}')H'}{H^2}. \quad (18)$$

Substituting (16) and (18) into (15), expanding the terms in the derivatives $(E\bar{u}' + F\bar{v}')'$ and $(F\bar{u}' + G\bar{v}')'$, and grouping terms involving $\bar{u}$, $\bar{v}$, and their derivatives gives

$$\bar{u}'' + \Gamma_{11}^1(\bar{u}')^2 + 2\Gamma_{12}^1\bar{u}'\bar{v}' + \Gamma_{22}^1(\bar{v}')^2 = 0, \quad \bar{v}'' + \Gamma_{11}^2(\bar{u}')^2 + 2\Gamma_{12}^2\bar{u}'\bar{v}' + \Gamma_{22}^2(\bar{v}')^2 = 0, \quad (19)$$

where $\bar{u}'' = \frac{d^2\bar{u}}{d\tau^2}$ and $\bar{v}'' = \frac{d^2\bar{v}}{d\tau^2}$. The pair of second order second-degree partial differential equations in (19) are known as the geodesic equations, and the Γ 's are known as the *Christoffel symbols*, which are given by

$$\begin{aligned} \Gamma_{11}^1 &= \frac{GE_{\bar{u}} - 2FF_{\bar{u}} + FE_{\bar{v}}}{2(EG - F^2)}, & \Gamma_{12}^1 &= \frac{GE_{\bar{v}} - FG_{\bar{u}}}{2(EG - F^2)}, & \Gamma_{22}^1 &= \frac{2GF_{\bar{v}} - GG_{\bar{u}} - FG_{\bar{v}}}{2(EG - F^2)}, \\ \Gamma_{11}^2 &= \frac{2EF_{\bar{u}} - EE_{\bar{v}} - FE_{\bar{u}}}{2(EG - F^2)}, & \Gamma_{12}^2 &= \frac{EG_{\bar{u}} - FE_{\bar{v}}}{2(EG - F^2)}, & \Gamma_{22}^2 &= \frac{EG_{\bar{v}} - 2FF_{\bar{v}} + FG_{\bar{u}}}{2(EG - F^2)}. \end{aligned} \quad (20)$$

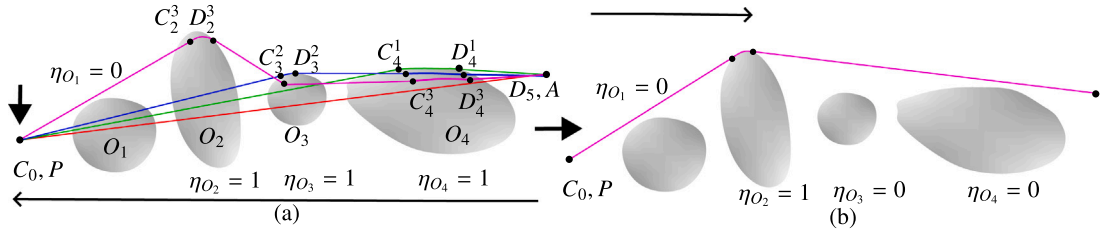


Fig. 5. Object-wrapping state detection. (a) Backward pass. (b) Forward pass.

- The endpoints of the cable are defined as $C_0 = P$ (the cable's origin) and $C_{n+1} = A$ (the fixed attachment point on the mobile-platform).
- $T(\tau) = \frac{d\alpha_{CD}}{d\tau}$ is the tangent vector along the cable curve, and $\hat{T}(\tau_{C_{\gamma,\zeta}})$, $\hat{T}(\tau_{D_{\gamma,\zeta}})$ are unit tangent vectors at the entry and exit points, $C_{\gamma,\zeta}$ and $D_{\gamma,\zeta}$, respectively, for object ζ .
- At $\mu_\zeta = \mu_\zeta^*$, $f_{C_\zeta}(\mu_\zeta^*) = 0$ and $f_{D_\zeta}(\mu_\zeta^*) = 0$, ensuring the tangential conditions are satisfied for all entry and exit points.

For a given pose q , solving (22) with μ_ζ^* and applying Proposition 4 determines the CWG $\bar{\alpha}_{CD_\zeta}^*(\tau)$, with its endpoints providing the unknown points C_ζ and D_ζ on object ζ . During simulations and experiments, this optimization is repeated for each pose in the trajectory $q(t)$. While solving the optimization problem for the current time-step, it is assumed that the initial guess for $\mathbf{M}$ is sufficiently close to its solution, which is achieved by using the solution from the previous time-step as the initial guess. This ensures efficiency and enables smooth transitions of CWGs across CDPR mobile-platform poses.

4. Cable-multi-object interference detection

In case of complex geometrical-shaped objects, the entry and exit points of cable wrapping on such objects are interdependent. A slight alteration in the location of these points can significantly affect the entire cable path. Additionally, in case of no-wrapping scenarios, the optimization problem in (22) is not required to be solved. Hence, this section discusses the procedure to detect whether cable γ in the CDPR is interfering, and then wrapping around multiple complex-shaped objects, indexed by ζ or not.

To clearly explain the workings of the detection algorithm within the CDPR environment, a scenario $k = 4$, and no wrapping on mobile-platform is considered (Fig. 5). For a given q , if interference is detected then object-wrapping state η_ζ , for $\zeta = 1, \dots, 4$ is given by (2). After detection, the condition of a cable path is characterized by a sequence of line segments and CWGs. These line segments are described by the entities that terminate each one, starting with the cable winch that is attached to the fixed frame and ending with the cable anchor affixed to the mobile-platform (line PA). The objects corresponds to and holds details about an individual object such as its geometry, its CWG model and its parent and child information, except for the initial and final objects, which symbolize the cable winch ($C_0 = P$) and mobile-platform ($D_5 = A$) ends. Starting from P and ending with A , the cable connection map (CCM) holds information regarding connection of the objects in the present iteration. Algorithm 1 provides the step-by-step procedure for cable-object interference detection, which involves backward and forward passes.

4.1. Backward pass

The objective of the backward pass is to determine the closest child object to the current object, such that the actual cable path wraps around the child object. Initially P is set as the parent object and A as the child.

In the first iteration, starting with an initial Cable Connection Map (CCM) of $\{P, A\}$ at the CDPR mobile-platform end ($D_5 = A$), the algorithm checks for intersections between line PA and object O_4 , as shown in Fig. 5(a). Upon detecting an intersection ($\eta_4 = 1$), the cable path is updated to $PC_4^1D_4^1A$ using (22), and the CCM is revised to $\{P, O_4, A\}$. In this configuration, P becomes the new parent of O_4 , and O_4 becomes the new parent of A .

In subsequent iterations, the algorithm extends this approach. From O_4 , it examines line PC_4^1 for intersections with O_3 , updating the cable path to $PC_3^2D_3^2C_4^1D_4^1A$ and the CCM to $\{P, O_3, O_4, A\}$, if $\eta_3 = 1$. The process continues with O_2 , updating the CCM to $\{P, O_2, O_3, O_4, A\}$ and deriving a new path $PC_2^3D_2^3C_3^2D_3^2C_4^1D_4^1A$. However, this path does not match the expected configuration of $PC_2^3D_2^3A$ (Fig. 5(b)), prompting a forward pass to correct the CCM to $\{P, O_2, A\}$. Finally, since O_1 does not intersect with PC_2^4 , O_2 is confirmed as the child of P .

4.2. Forward pass

The forward pass begins with O_2 as the parent and A as the child. To identify the next parent for the upcoming iteration, a backward pass is initiated with the initial line segment C_2A and CCM $\{P, O_2, A\}$. Examination reveals that C_2A does not intersect any objects from the perspective of A , thus the final CCM remains $\{P, O_2, A\}$. Consequently, the final cable path is confirmed as $\{P, O_2, A\}$ with $\eta_2 = 1$ and $n = 1$, as depicted in Fig. 5(b). Finally, in a similar manner, this algorithm can be easily extended to $k + 1$ objects.

which can be re-written as

$$d_\gamma(\mathbf{r}_{D_{\zeta_n} C_{\zeta_p}}) - d_\delta(\mathbf{r}_{E_{\zeta,\delta}}) = \mathbf{r}_{D_{\zeta_n} E_{\zeta,\delta}}, \quad (26)$$

such that if

$$0 < d_\gamma \leq 1, \text{ and } 0 \leq d_\delta \leq 1, \quad (27)$$

then (24) and (26) provide the necessary and sufficient conditions for the occurrence of the interference. Here, it has been assumed that $d_\gamma = 0$ cannot be reached since the cable segment beginning point D_{ζ_n} does not belong to the line spanned by the edge direction vector $\mathbf{r}_{D_{\zeta_n} C_{\zeta_p}}$ else practically there would be no collision. If (24) is satisfied then the least-squares solution [16] of (26) can be written as

$$\begin{bmatrix} d_\gamma \\ d_\delta \end{bmatrix} = \frac{1}{s} \begin{bmatrix} (\mathbf{r}_{EG_{\zeta,\delta}} \times \mathbf{n}_\delta)^T \\ (\mathbf{r}_{D_{\zeta_n} C_{\zeta_p}} \times \mathbf{n}_\delta)^T \end{bmatrix} s \mathbf{r}_{D_{\zeta_n} E_{\zeta,\delta}}, \text{ where } s = \left\| \mathbf{r}_{D_{\zeta_n} C_{\zeta_p}} \right\|_2^2 \left\| \mathbf{r}_{EG_{\zeta,\delta}} \right\|_2^2 - \left(\mathbf{r}_{D_{\zeta_n} C_{\zeta_p}}^T \mathbf{r}_{EG_{\zeta,\delta}} \right)^2. \quad (28)$$

Remark 3. Hence, cable-object interference with the δ^{th} object edge detection begins by evaluating (24). If this condition is met, the values of d_γ and d_δ can be determined using (28). Interference is confirmed when (27) holds true.

5. CWG-based IK

By implementing the CWG models obtained in Section 3, this section formulates the CWGIK model of a 6 DoF CDPR tailored for object wrapping, which encompasses obstacle-self, obstacle, self, and no-wrapping scenarios.

Let the 6 DoF CDPR mobile-platform's joint position is given by $\mathbf{q} = [\mathbf{q}_t^T, \mathbf{q}_o^T]^T \in \mathbb{R}^6$, where $\mathbf{q}_t = {}^O \mathbf{r}_{OG} \in \mathbb{R}^3$ provides the translation of the mobile-platform's Centre of Mass (CoM) and $\mathbf{q}_o \in \mathbb{R}^3$ are the Euler angles describing the orientation of the mobile-platform. The CWGIK problem deals with the determination of cable lengths $\mathbf{l} = [l_1, \dots, l_m]^T \in \mathbb{R}^m$ for various wrapping states for a given $\mathbf{q}$. Hence, CWGIK mapping from $\mathbf{q}$ to $\mathbf{l}$ can be referred to as a mapping function ξ , defined as $\xi: \mathbf{q} \rightarrow \mathbf{l}$. The CWGIK model can be written as

$$\mathbf{l} = \xi(\mathbf{q}), \quad (29)$$

where $\xi = [\xi_1, \dots, \xi_m]^T \in \mathbb{R}^m$. Taking time-derivative of (29) yields

$$\dot{\mathbf{l}} = \mathbf{J} \dot{\mathbf{q}}, \quad (30)$$

where

$$\mathbf{J} = \begin{bmatrix} \frac{\partial \xi_1}{\partial \mathbf{q}}^T & \dots & \frac{\partial \xi_m}{\partial \mathbf{q}}^T \end{bmatrix}^T \in \mathbb{R}^{m \times 6}$$

is the Jacobian matrix, where

$$\frac{\partial \xi_\gamma}{\partial \mathbf{q}} = \begin{bmatrix} \frac{\partial \xi_\gamma}{\partial q_1} & \dots & \frac{\partial \xi_\gamma}{\partial q_6} \end{bmatrix}^T \in \mathbb{R}^6.$$

In order to determine $\mathbf{J}$, let $\dot{\mathbf{p}} = [\dot{\mathbf{r}}_{OG}^T, \dot{\boldsymbol{\omega}}_G^T]^T \in \mathbb{R}^6$ denotes the *body twist vector*, containing the absolute velocity of the CoM $\dot{\mathbf{r}}_{OG}^T$ and the absolute angular velocity $\dot{\boldsymbol{\omega}}_G^T$ of the end-effector. The body twist vector $\dot{\mathbf{p}}$ can be related to the joint space velocity vector $\dot{\mathbf{q}}$ in the form of

$$\dot{\mathbf{p}} = \mathbf{S} \dot{\mathbf{q}}, \quad (31)$$

where $\mathbf{S} = \begin{bmatrix} {}^G \mathbf{R} & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & \mathbf{H} \end{bmatrix} \in \mathbb{R}^{6 \times 6}$, ${}^G \mathbf{R} \in \mathbb{R}^{3 \times 3}$ is a rotation matrix transforming a vector from frame $\{F_O\}$ to $\{F_G\}$, $\mathbf{0}_{3 \times 3}$ is 3×3 zero matrix, and $\mathbf{H} \in \mathbb{R}^{3 \times 3}$ provides mapping from Euler angle rates to body-axis rates. Likewise, the rate of change of cable lengths $\dot{\mathbf{l}}$ can also be related to the body twist vector $\dot{\mathbf{p}}$ in the form of

$$\dot{\mathbf{l}} = [\dot{l}_1, \dots, \dot{l}_m]^T = \mathbf{V} \dot{\mathbf{p}}, \quad (32)$$

where $\mathbf{V} \in \mathbb{R}^{m \times 6}$ relates the body twist vector $\dot{\mathbf{p}}$ to the rate change of cable lengths $\dot{\mathbf{l}}$. From (31), and (32), Eq. (30), can be re-written as

$$\dot{\mathbf{l}} = \mathbf{J} \dot{\mathbf{q}} = \mathbf{V} \mathbf{S} \dot{\mathbf{q}}, \quad (33)$$

where if matrix $\mathbf{S}$ is known then Jacobian matrix $\mathbf{J}$ can be determined by evaluating the matrix $\mathbf{V}$.

5.1. Rate of change of cable length

This section deals with the determination of the rate of change of cable γ length $\dot{l}_\gamma$ for object-wrapping and its various cases.

Theorem 1. During CDPR operation, if cable γ attachment points P and A are known, then the rate of change of cable γ length is given by

$$\dot{l}_\gamma = \left[G \hat{\mathbf{r}}_{C_n D_{n+1}}^T, \left(G \mathbf{r}_{GD_{n+1}} \times G \hat{\mathbf{r}}_{C_n D_{n+1}} \right)^T \right] \dot{\mathbf{p}}, \quad (34)$$

where

- $G \hat{\mathbf{r}}_{C_n D_{n+1}}$ represents the unit vector of $G \mathbf{r}_{C_n D_{n+1}}$, and
- point C_n and D_{n+1} can be determined by solving (22). $\circ$

Proof. From Fig. 2(a), the rate of change of cable γ 's length can be written as

$$\dot{l} = \sum_{\zeta=1}^n \left(\dot{l}_{C_{\zeta-1} D_\zeta} + \dot{l}_{CD_\zeta} \right) + \dot{l}_{C_n D_{n+1}} + \dot{l}_{AD_{n+1}}, \quad (35)$$

where the subscript γ has been removed from the right hand side terms since the proof is applicable for cables $\gamma = 1 \dots n$. Terms $\dot{l}_{C_{\zeta-1} D_\zeta} = \left\| \dot{\mathbf{r}}_{C_{\zeta-1} D_\zeta}^{S_\zeta} \right\|_2$ and $\dot{l}_{C_n D_{n+1}} = \left\| G \mathbf{r}_{C_n D_{n+1}} \right\|_2$ represent rate of change of straight cable length segments such that $\zeta = 1 \dots n$, and $\dot{l}_{CD_\zeta}$ and $\dot{l}_{AD_{n+1}}$ represent the obstacles and mobile-platform rate of change of geodesic cable length segments. Next, the rate of change of cable length of each segment is derived.

Straight Segment $C_{\zeta-1} D_\zeta$

The vector $\mathbf{r}_{C_{\zeta-1} D_\zeta}^{S_\zeta}$ can be written as

$$\mathbf{r}_{C_{\zeta-1} D_\zeta}^{S_\zeta} = -\mathbf{r}_{C_{\zeta-1}}^{S_\zeta} R^{S_{\zeta-1}} \mathbf{r}_{S_{\zeta-1} C_{\zeta-1}} + \mathbf{r}_{S_{\zeta-1}}^{S_\zeta} \mathbf{r}_{S_{\zeta-1} S_\zeta} + \mathbf{r}_{SD_\zeta}^{S_\zeta}. \quad (36)$$

Taking the time derivative of (36) gives

$$\dot{\mathbf{r}}_{C_{\zeta-1} D_\zeta}^{S_\zeta} = -\dot{\mathbf{r}}_{C_{\zeta-1}}^{S_\zeta} R^{S_{\zeta-1}} \mathbf{r}'_{S_{\zeta-1} C_{\zeta-1}} + \mathbf{r}_{S_{\zeta-1}}^{S_\zeta} \dot{\mathbf{r}}'_{SD_\zeta}, \quad (37)$$

which can be expanded as

$$\dot{\mathbf{r}}_{C_{\zeta-1} D_\zeta}^{S_\zeta} = \dot{\mathbf{r}}_{C_{\zeta-1}}^{S_\zeta} \boldsymbol{\omega}_{S_\zeta} \times \left(-\mathbf{r}_{C_{\zeta-1}}^{S_\zeta} \mathbf{r}_{S_{\zeta-1} C_{\zeta-1}} + \mathbf{r}_{SD_\zeta}^{S_\zeta} \right) - \mathbf{r}_{C_{\zeta-1}}^{S_\zeta} \dot{\mathbf{r}}'_{S_{\zeta-1} C_{\zeta-1}} + \mathbf{r}_{SD_\zeta}^{S_\zeta} \dot{\mathbf{r}}'_{SD_\zeta} = -\mathbf{r}_{C_{\zeta-1}}^{S_\zeta} \dot{\mathbf{r}}'_{S_{\zeta-1} C_{\zeta-1}} + \mathbf{r}_{SD_\zeta}^{S_\zeta} \dot{\mathbf{r}}'_{SD_\zeta}, \quad (38)$$

where $\dot{\mathbf{r}}_{C_{\zeta-1}}^{S_\zeta} \boldsymbol{\omega}_{S_\zeta} = \mathbf{0}$ since the obstacle is stationary with frame $\{F_{S_\zeta}\}$, and $\mathbf{r}_{SD_\zeta}^{S_\zeta} \dot{\mathbf{r}}'_{SD_\zeta} = \dot{\mathbf{r}}_{SD_\zeta}^{S_\zeta} \hat{\mathbf{i}} + \dot{\mathbf{r}}_{SD_\zeta}^{S_\zeta} \hat{\mathbf{j}} + \dot{\mathbf{r}}_{SD_\zeta}^{S_\zeta} \hat{\mathbf{k}}$ denotes the rate of change of $\mathbf{r}_{SD_\zeta}^{S_\zeta}$ as observed from $\{F_{S_\zeta}\}$, where $(\dot{\mathbf{r}}_{SD_\zeta}^{S_\zeta} \hat{\mathbf{i}}, \dot{\mathbf{r}}_{SD_\zeta}^{S_\zeta} \hat{\mathbf{j}}, \dot{\mathbf{r}}_{SD_\zeta}^{S_\zeta} \hat{\mathbf{k}})$ are unit vectors along the axes of frame $\{F_{S_\zeta}\}$. Next, the time derivative of the straight cable length $\dot{l}_{C_{\zeta-1} D_\zeta}$ can be expressed as

$$\dot{l}_{C_{\zeta-1} D_\zeta} = \left\| \dot{\mathbf{r}}_{C_{\zeta-1} D_\zeta}^{S_\zeta} \right\|_2 = \dot{\mathbf{r}}_{C_{\zeta-1} D_\zeta}^{S_\zeta} \cdot \dot{\mathbf{r}}_{C_{\zeta-1} D_\zeta}^{S_\zeta}, \quad (39)$$

where $\dot{\mathbf{r}}_{C_{\zeta-1} D_\zeta}^{S_\zeta}$ represents the unit vector of $\mathbf{r}_{C_{\zeta-1} D_\zeta}^{S_\zeta}$. Substituting (38) in (39) yields

$$\dot{l}_{C_{\zeta-1} D_\zeta} = - \left(\mathbf{r}_{C_{\zeta-1}}^{S_\zeta} R^{S_{\zeta-1}} \hat{\mathbf{r}}_{C_{\zeta-1} D_\zeta} \right) \cdot \left(\mathbf{r}_{C_{\zeta-1}}^{S_\zeta} R^{S_{\zeta-1}} \dot{\mathbf{r}}'_{S_{\zeta-1} C_{\zeta-1}} \right) + \dot{\mathbf{r}}_{C_{\zeta-1} D_\zeta}^{S_\zeta} \cdot \mathbf{r}_{SD_\zeta}^{S_\zeta} \dot{\mathbf{r}}'_{SD_\zeta}, \quad (40)$$

can be simplified to

$$\dot{l}_{C_{\zeta-1} D_\zeta} = -\mathbf{r}_{C_{\zeta-1}}^{S_\zeta} \hat{\mathbf{r}}_{C_{\zeta-1} D_\zeta} \cdot \mathbf{r}_{S_{\zeta-1}}^{S_\zeta} \dot{\mathbf{r}}'_{S_{\zeta-1} C_{\zeta-1}} + \dot{\mathbf{r}}_{C_{\zeta-1} D_\zeta}^{S_\zeta} \cdot \mathbf{r}_{SD_\zeta}^{S_\zeta} \dot{\mathbf{r}}'_{SD_\zeta}, \quad (41)$$

where

$$\mathbf{r}_{C_{\zeta-1}}^{S_\zeta} \hat{\mathbf{r}}_{C_{\zeta-1} D_\zeta} = - \mathbf{r}_{C_{\zeta-1}}^{S_\zeta} \hat{\mathbf{T}}(\tau) \Big|_{\tau_{C_{\zeta-1}}(t)}, \quad (42)$$

$$\mathbf{r}_{C_{\zeta-1}}^{S_\zeta} \hat{\mathbf{r}}_{C_{\zeta-1} D_\zeta} = - \mathbf{r}_{C_{\zeta-1}}^{S_\zeta} \hat{\mathbf{T}}(\tau) \Big|_{\tau_{D_\zeta}(t)}, \quad (43)$$

$$\mathbf{r}_{S_{\zeta-1}}^{S_\zeta} \dot{\mathbf{r}}'_{S_{\zeta-1} C_{\zeta-1}} = \frac{d\tau_{C_{\zeta-1}}(t)}{dt} \mathbf{r}_{S_{\zeta-1}}^{S_\zeta} \mathbf{T}(\tau) \Big|_{\tau_{C_{\zeta-1}}(t)}$$

and

$$\mathbf{r}_{SD_\zeta}^{S_\zeta} \dot{\mathbf{r}}'_{SD_\zeta} = \frac{d\tau_{D_\zeta}(t)}{dt} \mathbf{r}_{SD_\zeta}^{S_\zeta} \mathbf{T}(\tau(t)) \Big|_{\tau_{D_\zeta}(t)}, \quad (44)$$

respectively, where

$$\mathbf{r}_{C_{\zeta-1}}^{S_\zeta} \hat{\mathbf{T}}(\tau) = \frac{1}{\left\| \frac{d\mathbf{r}_{C_{\zeta-1}}^{S_\zeta} \hat{\mathbf{T}}(\tau)}{d\tau} \right\|_2} \frac{d\mathbf{r}_{C_{\zeta-1}}^{S_\zeta} \hat{\mathbf{T}}(\tau)}{d\tau}, \quad (45)$$

is the unit tangent vector of the CWG at τ on obstacle $\zeta - 1$, represented with respect to frame $F_{S_{\zeta-1}}$. Substituting (42), (43) and (44) in (41) gives

$$\dot{l}_{C_{\zeta-1}D_{\zeta}} = \frac{d\tau_{C_{\zeta-1}}(t)}{dt} \left\| S_{\zeta-1}^T T(\tau) \right\|_2 \Big|_{\tau_{C_{\zeta-1}}(t)} - \frac{d\tau_{D_{\zeta}}(t)}{dt} \left\| S_{\zeta}^T T(\tau) \right\|_2 \Big|_{\tau_{D_{\zeta}}(t)}. \quad (46)$$

CWG Segment CD_{ζ}

Next, CWG length $\dot{l}_{CD_{\zeta}}$ from $\tau = \tau_{C_{\zeta}}(t)$ to $\tau = \tau_{D_{\zeta}}(t)$ can be written as

$$l_{CD_{\zeta}} = \int_{\tau_{C_{\zeta}}(t)}^{\tau_{D_{\zeta}}(t)} \left\| \frac{d\alpha_{CD_{\zeta}}}{d\tau} \right\|_2 d\tau \quad (47)$$

By applying *Leibniz integral rule*, time derivative of (47) can be written as

$$\dot{l}_{CD_{\zeta}} = \frac{d\tau_{D_{\zeta}}(t)}{dt} \left\| S_{\zeta}^T T(\tau) \right\|_2 \Big|_{\tau_{D_{\zeta}}(t)} - \frac{d\tau_{C_{\zeta}}(t)}{dt} \left\| S_{\zeta}^T T(\tau) \right\|_2 \Big|_{\tau_{C_{\zeta}}(t)}, \quad (48)$$

where the integral part in the Leibniz integral rule is zero since $\alpha_{CD_{\zeta}}$ is solely a function of τ .

Straight Segment $\dot{l}_{C_n D_{n+1}}$

The direction vector ${}^G \mathbf{r}_{C_n D_{n+1}}$ can be written as

$${}^S \mathbf{r}_{C_n D_{n+1}} = {}^S \mathbf{r}_{OG} + {}^S \mathbf{r}_{GD_{n+1}} - {}^S \mathbf{r}_{OS_n} - {}^S \mathbf{r}_{SC_n}. \quad (49)$$

Taking the time derivative of (49) and pre-multiplying both sides with ${}^G R$ gives

$${}^G \dot{\mathbf{r}}_{C_n D_{n+1}} = {}^G \dot{\mathbf{r}}_{OG} + {}^G \omega_G \times {}^G \mathbf{r}_{GD_{n+1}} + {}^G \mathbf{r}'_{GD_{n+1}} - {}^G R {}^S \mathbf{r}'_{SC_n}, \quad (50)$$

where ${}^O \mathbf{r}_{OS_n} = \mathbf{0}$ since ${}^O \mathbf{r}_{OS_n}$ is fixed to the inertial frame. By using (50), the time derivative of the straight cable length $\dot{l}_{C_n D_{n+1}}$ can be expressed as

$$\dot{l}_{C_n D_{n+1}} = {}^G \hat{\mathbf{r}}_{C_n D_{n+1}}^T {}^G \dot{\mathbf{r}}_{C_n D_{n+1}} = \left[{}^G \hat{\mathbf{r}}_{C_n D_{n+1}}^T \left({}^G \mathbf{r}_{GD_{n+1}} \times {}^G \hat{\mathbf{r}}_{C_n D_{n+1}} \right)^T \right] \begin{bmatrix} {}^G \mathbf{r}_{OG} \\ {}^G \omega_G \end{bmatrix} + \left({}^G \hat{\mathbf{r}}_{C_n D_{n+1}} \right) \cdot \left({}^G \mathbf{r}'_{GD_{n+1}} \right) - \left({}^G R {}^S \hat{\mathbf{r}}_{C_n D_{n+1}} \right) \cdot \left({}^G R {}^S \mathbf{r}'_{SC_n} \right). \quad (51)$$

which can be simplified as

$$\dot{l}_{C_n D_{n+1}} = \left[{}^G \hat{\mathbf{r}}_{C_n D_{n+1}}^T \left({}^G \mathbf{r}_{GD_{n+1}} \times {}^G \hat{\mathbf{r}}_{C_n D_{n+1}} \right)^T \right] \dot{p} - \frac{d\tau_{D_{n+1}}(t)}{dt} \left\| {}^G T(\tau) \right\|_2 \Big|_{\tau_{D_{n+1}}(t)} + \frac{d\tau_{C_n}(t)}{dt} \left\| {}^S T(\tau) \right\|_2 \Big|_{\tau_{C_n}(t)}, \quad (52)$$

where ${}^G R^T {}^G R = I_{3 \times 3}$, ${}^G \hat{\mathbf{r}}_{C_n D_{n+1}} \cdot {}^G \mathbf{r}'_{GD_{n+1}} = -\frac{d\tau_{D_{n+1}}(t)}{dt} \left\| {}^G T(\tau) \right\|_2 \Big|_{\tau_{D_{n+1}}(t)}$, and ${}^S \hat{\mathbf{r}}_{C_n D_{n+1}} \cdot {}^S \mathbf{r}'_{SC_n} = -\frac{d\tau_{C_n}(t)}{dt} \left\| {}^S T(\tau) \right\|_2 \Big|_{\tau_{C_n}(t)}$.

CWG Segment AD_{n+1}

By applying *Leibniz integral rule*, $\dot{l}_{AD_{n+1}}$ can be written as

$$\dot{l}_{AD_{n+1}} = \frac{d\tau_{D_{n+1}}(t)}{dt} \left\| {}^G T(\tau) \right\|_2 \Big|_{\tau_{D_{n+1}}(t)} \quad (53)$$

Rate of change of cable length $\dot{l}_{\gamma}$

Adding (46), (48), (52) and (53) yields (34). $\square$

Proposition 5. If there are k obstacles and a mobile-platform (a total of $k + 1$ objects), then by applying *Theorem 1*, the rate of change of the length of cable γ , as given by (34), can be deduced for various cable wrapping scenarios by determining the corresponding C_n (cable entry points) and D_n (cable exit points) under the following wrapping conditions

$$C_n = \begin{cases} \neq D_n & \text{if multi-wrapping } \eta_n = 1, \eta_{n+1} = 1, \\ \neq D_n & \text{if obstacle-wrapping } \eta_n = 1, \eta_{n+1} = 0, \\ C_0 = P & \text{if self-wrapping } n = 0, \eta_1 = 1, \\ C_0 = P & \text{if no-wrapping } n = 0, \eta_1 = 0, \end{cases} \quad D_{n+1} = \begin{cases} \neq C_{n+1} & \text{if multi-wrapping } \eta_n = 1, \eta_{n+1} = 1, \\ C_{n+1} = A & \text{if obstacle-wrapping } \eta_n = 1, \eta_{n+1} = 0, \\ D_1 \neq C_1 & \text{if self-wrapping } n = 0, \eta_1 = 1, \\ D_1 = A & \text{if no-wrapping } n = 0, \eta_1 = 0, \end{cases} \quad (54)$$

where object wrapping state η_{ζ} for $\zeta = 1, \dots, k + 1$ can be determined by the cable-object interference detection algorithm (Algorithm 1) discussed in Section 4.

5.2. Obtaining the Jacobian matrix J

Obtaining (34) for $\gamma = 1, \dots, m$ allows the time derivative of the cable length vector to be expressed in the form of $\dot{l} = V \dot{p}$ as per (32), where

$$V = \begin{bmatrix} G \hat{r}_{C_{1,n}D_{1,n+1}}^T & \left(G \mathbf{r}_{GD_{1,n+1}} \times G \hat{r}_{C_{1,n}D_{1,n+1}} \right)^T \\ \vdots & \\ G \hat{r}_{C_{m,n}D_{m,n+1}}^T & \left(G \mathbf{r}_{GD_{m,n+1}} \times G \hat{r}_{C_{m,n}D_{m,n+1}} \right)^T \end{bmatrix}. \quad (55)$$

Furthermore as per (33) the change in cable length vector $\dot{l}$ can be related to the joint space velocity vector $\dot{q}$ linearly in the form of $\dot{l} = J \dot{q}$. Hence, by post-multiplying (55) with S on both sides, the Jacobian matrix J can be obtained as

$$J = VS. \quad (56)$$

6. CWG-based ID and cable-surface friction modeling

The dynamics of the CDPR system with 6 DoF actuated by m cables can be modeled by formulating its *Equation of Motion (EoM)*, which can be expressed as

$$M(q)\ddot{q} + C(q, \dot{q}) + G(q) = -J^T f, \quad (57)$$

where $M(q)$, $C(q, \dot{q})$, and $G(q)$ are the *mass inertia matrix*, the *centrifugal and Coriolis components*, and the *gravitational components* of the system dynamics, respectively. The *cable wrench vector* is denoted by $-J^T f$, where $-J^T \in \mathbb{R}^{6 \times m}$ represents the direction of the forces and moments generated by the actuated cables, and $f = [f_1, \dots, f_m]^T$ is the cable force vector containing the magnitudes of the tension for individual cables, where f_γ is the tension in cable γ . As discussed in Chapter 3 of Lau's thesis [31], the EoM can be derived by calculating the resultant forces and moments about the center of mass.

For a predefined trajectory $q(t)$ of a CDPR system, the dynamics equation referenced in (57) can be rewritten as a linear system of equations

$$-J^T f = b, \quad (58)$$

where the vector $b = M(q)\ddot{q} + C(q, \dot{q}) + G(q)$. The CWGID involves calculating the cable tension vector f , which must adhere to the cable tension bounds given by $f_{lb} \leq f \leq f_{ub}$ and simultaneously satisfying (58). For cable systems that are completely and redundantly restrained, there is an infinite array of solutions to the CWGID problem. This challenge can be addressed by formulating an objective function that is based on the cable tensions. In this paper, the incorporation of cable wrapping in the CWGID problem of the CDPR system will be demonstrated by the following objective function

$$\begin{aligned} f^* &= \underset{f}{\operatorname{argmin}} f^T P f \\ \text{subject to } & M(q)\ddot{q} + C(q, \dot{q}) + G(q) = -J^T f, \\ & f_{lb} \leq f \leq f_{ub}, \end{aligned} \quad (59)$$

where P is a positive definite matrix.

During object wrapping, when a CWG moves on a surface, such as a NURBS surface, friction between the cable and the surface affects the tension distribution along the cable. The friction causes the tension to decrease along the cable as it moves, depending on the coefficient of friction μ and curvature κ ; the higher the κ , the greater the normal force, which enhances the frictional force. Hence, the tension at the motor side $f_{D_{\gamma,1}}$ must be greater than the tension at the far end $f_{C_{\gamma,n+1}}$ (Fig. 2), because friction reduces the tension along the cable as it wraps around the objects. This section explores the relationship between the tension at different points along the cable, accounting for friction and curvature, in the context of CWG on the NURBS surface.

Theorem 2. In the CDPR operation at a given pose q , let a cable γ wrap geodesically around $n+1$ objects (n obstacles and mobile-platform, Fig. 2). Assuming each object $\zeta = 1, \dots, n+1$ has an isotropic friction coefficient μ_ζ , and the cable wraps around it with a curvature $\kappa_{\gamma,\zeta}(s)$ over an arc length s from 0 to $s_{D_{\gamma,\zeta}}$. The relationship between the required motor side cable tension $f_{D_{\gamma,1}}$ with the CWGID calculated cable tension $f_{C_{\gamma,n+1}} = f_\gamma^*$ (tension without friction, Eq. (59)) is given by

$$f_{D_{\gamma,1}} = f_{C_{\gamma,n+1}} \prod_{\zeta=1}^{n+1} \left(\exp \left(\mu_\zeta \int_{s=0}^{s_{D_{\gamma,\zeta}}} \kappa_{\gamma,\zeta}(s') ds' \right) \right) \quad (60)$$

Proof. The frictional force $dF_{\gamma,\zeta}$ experienced by an infinitesimal segment ds of the cable γ as it wraps geodesically around object ζ is given by

$$dF_{\gamma,\zeta} = \mu_\zeta N_{\gamma,\zeta} ds, \quad (61)$$

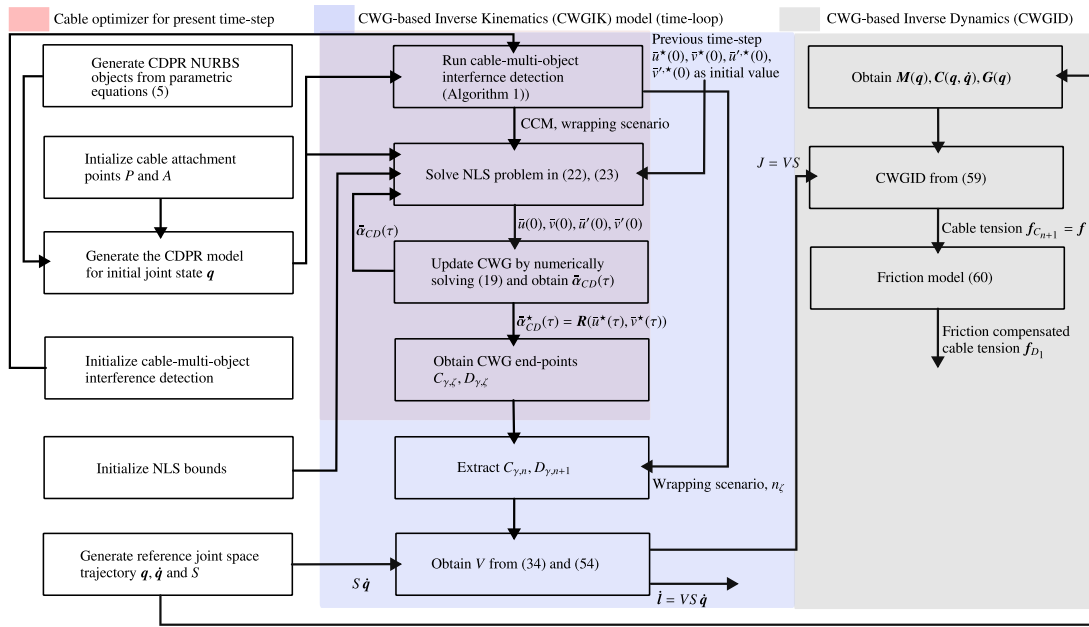


Fig. 7. Complete CDPR kinematic and dynamic framework with cable-object wrapping.

where $N_{\gamma,\zeta}$ is the normal force per unit length acting on the cable due to its curvature over object ζ , and is given by

$$N_{\gamma,\zeta} = f_{\gamma,\zeta}(s) \kappa_{\gamma,\zeta}(s), \quad (62)$$

where $f_{\gamma,\zeta}(s)$ is the cable tension at position s on object ζ . The differential change in tension along the cable from $C_{\gamma,\zeta}$ (low tension) to $D_{\gamma,\zeta}$ (high tension) due to friction as it wraps around object ζ is then

$$df_{\gamma,\zeta} = dF_{\gamma,\zeta} = \mu_{\zeta} N_{\gamma,\zeta} ds = \mu_{\zeta} f_{\gamma,\zeta}(s) \kappa_{\gamma,\zeta}(s) ds. \quad (63)$$

Rewriting (63)

$$\frac{df_{\gamma,\zeta}}{f_{\gamma,\zeta}(s)} = \mu_{\zeta} \kappa_{\gamma,\zeta}(s) ds. \quad (64)$$

Integrating both sides of (64) from $s = 0$ to $s_{D_{\gamma,\zeta}}$, simplifying and re-arranging it obtains

$$f_{D_{\gamma,\zeta}} = f_{D_{\gamma,\zeta+1}} \exp \left(\mu_{\zeta} \int_0^{s_{D_{\gamma,\zeta}}} \kappa_{\gamma,\zeta}(s') ds' \right), \quad (65)$$

where $f_{\gamma,\zeta}(0) = f_{C_{\gamma,\zeta}} = f_{D_{\gamma,\zeta+1}}$ and $f_{\gamma,\zeta}(s_{D_{\gamma,\zeta}}) = f_{D_{\gamma,\zeta}}$. Starting from the desired end tension (after all wrappings) at object $\zeta = n + 1$, given by $f_{D_{\gamma,n+2}} = f_{C_{\gamma,n+1}} = f_{\gamma}^*$, (65) can be recursively written as (60) for $\zeta = n + 1, \dots, 1$. $\square$

Fig. 7 provides the complete kinematic and dynamic modeling framework to operate a CDPR with cable-object wrapping. The left section focuses on initializing the CDPR model, including generating cable attachment points, NURBS parametric equations, and trajectory references. The middle section explains the CWGIK, which involves solving a Nonlinear Least Squares (NLS) optimization problem to compute CWGs and extract wrapping endpoints for multi-object cable wrapping. The right section addresses the CWGID component, which computes cable tensions, incorporates friction compensation. The CWGID simulation results illustrating the cable force without friction consideration and the effects of frictional compensation are presented in Section 7.5.

7. Simulation results

To demonstrate the framework's versatility, CWG models were generated for various complex shapes and used for cable-object interference detection, optimizing cable paths for different object configurations. Simulations for the entire framework were conducted using a cone-frustum-shaped mobile-platform (mobile-platform's main body) and a teapot-shaped obstacle.

7.1. Object surface description

The teapot obstacle is composed of three distinct objects: Cover O_1 , Body O_2 , and Handle O_3 (Fig. 9). The parametric equations for these teapot objects (O_1 , O_2 , and O_3) are obtained by generating NURBS surfaces as per (6), Section 3.1. The parametric surface of

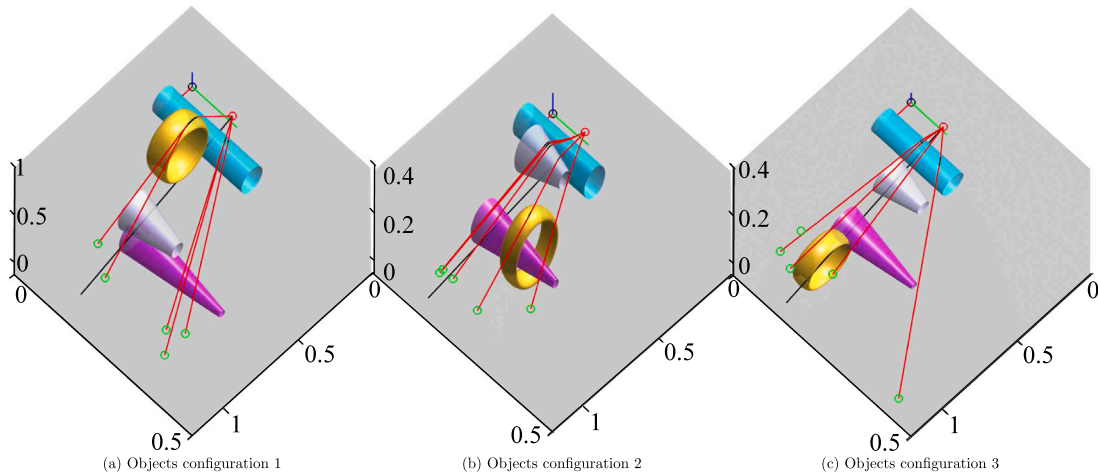


Fig. 8. Results of cable-object Interference detection algorithm with four objects, and three distinct object location configurations.

the cone-frustum is generated from $[\tan(\beta)(u+u_0)\cos(v), u, \tan(\beta)(u+u_0)\sin(v)]^T$, where the cone-apex angle β and initial u displacement are set to -0.1489 rad and -0.4 m, respectively. The non-zero Christoffel symbols of the cone-frustum were computed using (20), which yields

$$F_{22}^1 = -\frac{\tan^2(\beta)\bar{u}}{1 + \tan^2(\beta)}, \quad F_{12}^2 = \frac{1}{\bar{u}}. \quad (66)$$

From (66), it can be seen that the Christoffel symbols of the cone-frustum are highly nonlinear compared to the cylindrical-shaped mobile-platform implemented in [12,22,23]. Thus, the solution of the geodesic equations for complex geometrically-shaped surfaces such as NURBS surfaces cannot be obtained analytically, unlike with cylindrical surfaces. In this study, numerical solutions for the Christoffel symbols of complex geometric surfaces, such as NURBS surfaces, were obtained.

7.2. Cable-object interference detection

The cable-object interference detection algorithm (Section 4) for multiple objects was validated using four distinct surfaces: (a) one cylinder, (b) two cone-frustums, and (c) one torus. The algorithm was tested across several object location configurations, with three representative examples shown in Fig. 8(a)–(c). For each configuration, the algorithm was executed with five different randomized y , and z axes values of A . The successful implementation of the algorithm resulted in appropriate wrapping configurations between fixed point P and variable point A , demonstrating its effectiveness in handling cable-object interference for complex object configurations.

7.3. BMWrapArm description

Simulation results on a 3-DoF CDPR ($n = 3$), known as the *BMWrapArm* (Fig. 9), are presented in this section to demonstrate the capability of the proposed CWGIK and CWGID framework. The *BMWrapArm* is configured with a spherical joint at its center, located at point O . The joint's configuration in the joint space is represented by the vector $\mathbf{q} = [q_1, q_2, q_3]^T$, where q_1 , q_2 , and q_3 are the Euler angles in the xyz sequence relative to the inertial frame $\{F_O\}$, whose origin is also at point O (Fig. 9). At O , frame $\{F_E\}$ is located, which shares its origin with $\{F_O\}$ and whose orientation is determined by the Euler angles in $\mathbf{q}$. The mobile-platform includes a cylindrical-shaped bar attached to the spherical joint, onto which the cone-frustum is mounted.

The base of the cone-frustum (mobile-platform's main body) is located at 0.068 m from the center of the spherical joint along the cone-frustum axis. The CoM of the cone-frustum mobile-platform is located at ${}^E\mathbf{r}_{OG} = [0, 0.147, 0]^T$ m. The origin of frame $\{F_G\}$ is fixed to the CoM of the *BMWrapArm* mobile-platform and its orientation is defined by $\mathbf{q}$. The base center of the teapot body is located at ${}^O\mathbf{r}_{OS} = [0.135, 0.15, 0]^T$ m, which is the origin of the frame $\{F_S\}$. The orientation of $\{F_S\}$ is aligned along $\{F_O\}$. In this case study, $\{F_S\}$ is considered as obstacle body frame for all obstacles O_1 , O_2 , and O_3 , however individual frames can also be considered. Table 1 provides other specifications of the obstacle and mobile-platform surfaces. The joint space is actuated by $m = 4$ cables ($\gamma = 1, \dots, 4$), which are attached according to the attachment points given in the fourth row of Table 1 such that the cables were pulled or released by sending length commands $\mathbf{l} = [l_1, l_2, l_3, l_4]^T \in \mathbb{R}^4$, obtained from the CWGIK simulations, discussed in Section 7.4.

Software description: In this work, various NURBS surfaces were generated by using Rhino 8, and after that the NURBS parameters are saved in an OBJ file. The CWGIK and CWGID simulations were performed with the open-source CASPR [32] software on a computer with an Intel Core i9-11900K CPU @ 3.50 GHz and 32.0 GB of RAM, using MATLAB R2022a (64-bit). The NLS optimization problems in (22) is solved by using the *fmincon interior-point* algorithm.

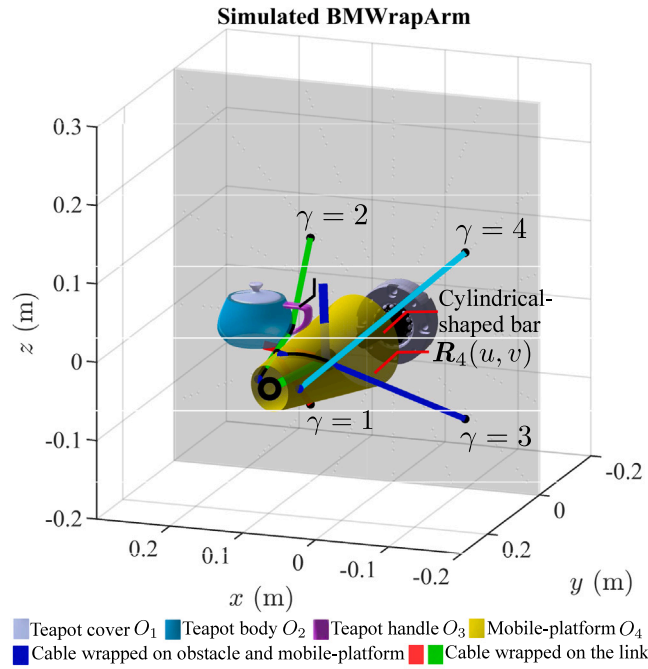


Fig. 9. Illustration of BMWrapArm simulation. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1

Parameter values used for model simulation.

Obstacle: Teapot	Mobile-platform: Cone-frustum		
Frame location and cable attachment points			
$\vec{OP}_1 = [0.1414, 0.0420, -0.1414]^T$ m, $\vec{OP}_3 = [-0.1414, 0.0420, -0.1414]^T$ m, $u_{A_1} = 0.34, u_{A_2} = 0.18$, m $v_{A_1} = 1.52, v_{A_2} = 0.00$, rad $\vec{EA}_1 = [0.0021, 0.1700, 0.0344]^T$ m, $\vec{EA}_3 = [-0.0021, 0.1700, 0.0344]^T$ m,	$\vec{OP}_2 = [0.1414, 0.0420, 0.1414]^T$ m $\vec{OP}_4 = [0.1414, 0.0420, -0.1414]^T$ m $u_{A_3} = 0.17, u_{A_4} = 0.18$ m $v_{A_3} = 4.65, v_{A_4} = 3.14$ rad $\vec{EA}_2 = [0.0330, 0.1800, 0]^T$ m $\vec{EA}_4 = [-0.0330, 0.1800, 0]^T$ m		
NLS bounds			
$\gamma = 2$: $O_1 : \mu_{1,l} = [0, 0.5, 0, -0.1]^T$, $\mu_{1,u} = [1, 1, 0.5, 0]^T$ $O_2 : \mu_{2,l} = [0, 0, -0.1, -0.1]^T$, $\mu_{2,u} = [1, 1, 0, 0]^T$ $O_3 : \mu_{3,l} = [0, 0.5, 0, -0.1]^T$, $\mu_{3,u} = [1, 1, 0.5, 0]^T$	$O_4, \gamma = 1: \mu_{4,l} = [-5, -0.5]^T$, $\mu_{4,u} = [1, 0.2]^T$ $O_4, \gamma = 2: \mu_{4,l} = [-1, -0.1]^T$, $\mu_{4,u} = [3, -0.0001]^T$ $O_4, \gamma = 3: \mu_{4,l} = [-5, -0.5]^T$, $\mu_{4,u} = [1, 0.2]^T$ $O_4, \gamma = 4: \mu_{4,l} = [-1, -0.1]^T$, $\mu_{4,u} = [3, -0.0001]^T$		
Obstacle teapot NURBS specifications			
	Teapot cover:	Teapot body:	Teapot handle:
Number of control points $(n_u + 1, n_v + 1)$	(13, 7)	(13, 7)	(7, 7)
Degree (p_u, p_v)	(3, 3)	(3, 3)	(3, 3)
Multiplicity (U, V)	{4, 3, 3, 3, 4}, {4, 3, 4}	{4, 3, 3, 3, 4}, {4, 3, 4}	{4, 1, 1, 1, 4}, {4, 1, 1, 1, 4}
Knot vector (U)	[0, 0, 0, 0, 0.25, 0.25, 0.25, 0.5, 0.5, 0.5, 0.5, 0.75, 0.75, 0.75, 1, 1, 1, 1]	[0, 0, 0, 0, 0.25, 0.25, 0.25, 0.5, 0.5, 0.5, 0.5, 0.75, 0.75, 0.75, 1, 1, 1, 1]	[0, 0, 0, 0, 0.25, 0.5, 0.75, 1, 1, 1, 1]
Knot vector (V)	[0, 0, 0, 0, 0.5, 0.5, 0.5, 1, 1, 1, 1]	[0, 0, 0, 0, 0.5, 0.5, 0.5, 1, 1, 1, 1]	[0, 0, 0, 0, 0.25, 0.5, 0.75, 1, 1, 1, 1]

7.4. CWGIK simulation

Trajectory simulation: By using the bounds given by the last row of Table 1 and a sampling time of $T_s = 0.05$ s (20 Hz), the NLS problem given by (22) was solved for a cone-frustum shaped mobile-platform and a teapot shaped obstacle to obtain the CWGIK-based cable lengths for a yaw joint-space trajectory $q(t)$ (Fig. 10(a)). The BMWrapArm CWGIK simulation commenced from an initial

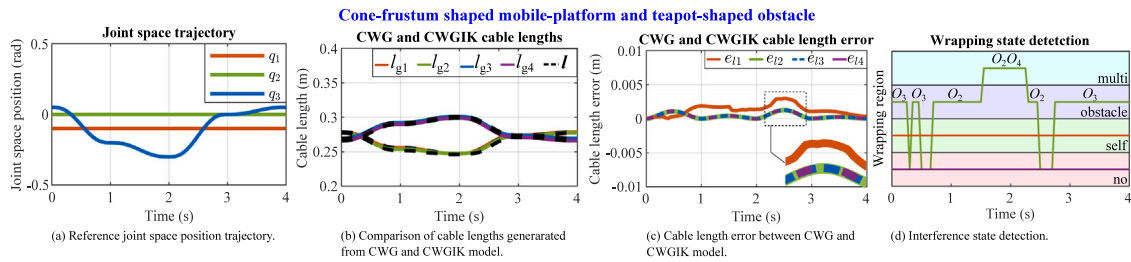


Fig. 10. Simulation results showing comparison between cable length obtained from the CWG model and the CWGIK.

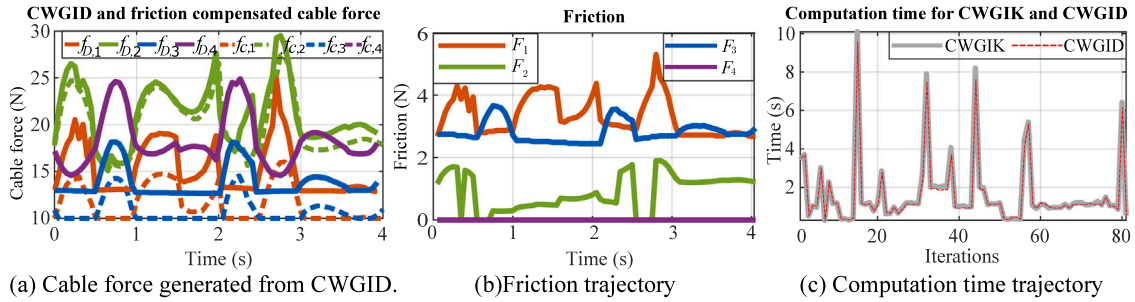


Fig. 11. CWGID simulation results.

joint position of $q(0) = [-0.1, 0.0, 0.05]^T$ rad. For this initial configuration, the NLS bounds, given by (22), were initially tuned to perform the NLS optimization to obtain the optimum cable lengths. To validate the accuracy of the CWG model, Fig. 10(b) shows the comparison of the CWG lengths $l_g = [l_{g1}, \dots, l_{g4}]^T$ obtained from the CWG model directly, discussed in Sections Section 3, and CWGIK cable lengths l obtained from the CWGIK, discussed in Section 5. Fig. 10(c) provides the cable length error plot between l_g and l .

Wrapping state detection: Within the CWGIK model, the object wrapping state detection algorithm, as discussed in Section 4 and detailed in Algorithm 1, played a crucial role in the dynamic simulation of the BMWrapArm. Within the first 2 s of the joint-space trajectory (Fig. 10(a)), cable 2 (green colored cable, Fig. 9) was found to be wrapping around the teapot handle (O_3) at the start of the trajectory, as shown in Fig. 10(d). From this initial wrapping around O_3 , the cable transitioned into a no-wrapping state, followed by wrapping around the teapot body (O_2), which triggered the obstacle-wrapping state. Subsequently, the cable entered a multi-wrapping state, encompassing both the teapot body (O_2) and the cone-frustum mobile-platform (O_1), indicating a much more complex interaction of the cable with the objects. From $t = 2$ s onward, the wrapping state trajectory followed a symmetrical mirrored path, reflecting the movements observed during the initial phase, and highlighting the repeatability of the detection.

7.5. CWGID simulation

Initialization: For the CWGID simulation, a mass of $m = 1$ kg was considered for the cone-frustum mobile-platform. The moment of inertia tensor matrix of the mobile-platform about its center of mass at point G is defined as

$$\mathbf{I}_G = \text{diag}(I_{G_x}, I_{G_y}, I_{G_z}), \quad (67)$$

where the components calculated as $I_{G_x} = I_{G_z} = 0.974 \text{ kg/m}^2$, and $I_{G_y} = 0.119 \text{ kg/m}^2$. Given the yaw joint-space trajectory $q(t)$, the BMWrapArm CWGID cable force trajectory was obtained by solving the redundancy resolution problem formulated in (59) subject to the equations of motion (EoM), with force bounds $f_{lb} = [10, 10, 10]^T \text{ N}$ and $f_{ub} = [80, 80, 80]^T \text{ N}$, and $P = I_{4 \times 4}$.

Friction compensated ID simulation: The trajectory is shown in Fig. 11(a) by dashed lines, where the sharp corners are primarily attributed to the rapid changes in the position of point $C_{2,m}$ on cable 2 in $f_{C_{2,m}D_{2,m+1}}^T$ in (55) on the surfaces of O_2 and O_3 during the BMWrapArm motion. The solid line trajectory in Fig. 11(a) represents the friction compensated cable force and Fig. 11(b) illustrates the cable-surface friction computed using (60). Finally, Fig. 11(c) displays the computation time required for each iteration of the CWGIK and CWGID simulations. The computation time per iteration is usually higher, up to $t = 8$ s, when solving multi-wrapping problems with complex geometrical surfaces.

The simulation results from the CWGIK and CWGID models with object detection algorithm demonstrate their ability to accurately calculate cable lengths and forces, respectively, in the presence of various cable wrapping scenarios for complex-shaped obstacles and mobile-platform surfaces. This underscores the models' effectiveness in managing intricate wrapping scenarios.

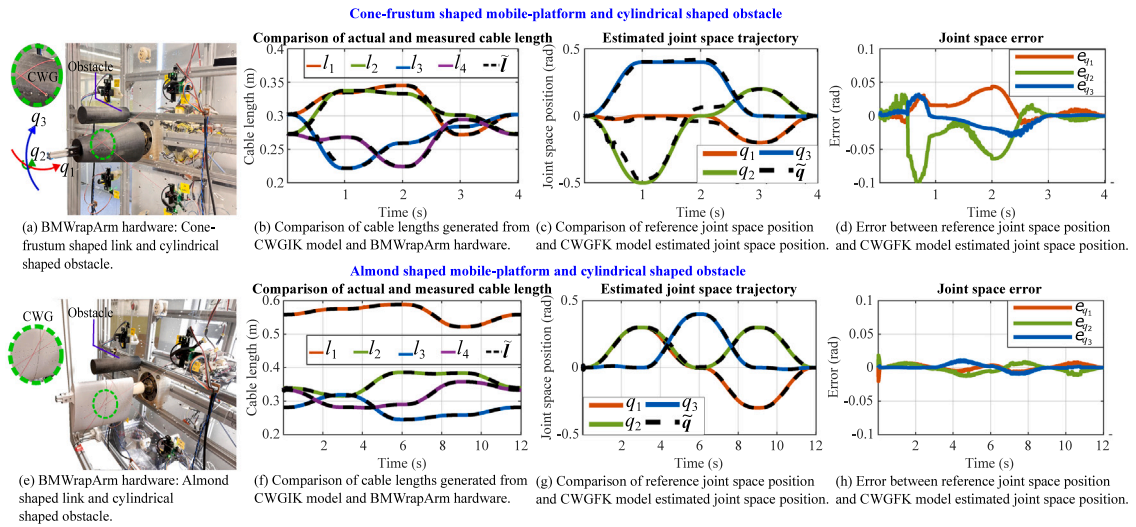


Fig. 12. BMWrapArm hardware setup and experimental results showing comparison between the CWGIK and BMWrapArm measured cable length, and reference and CWGFK estimated joint-space trajectory.

Table 2

Derived Christoffel's symbols of the chosen BMWrapArm mobile-platforms and obstacles.

Surface type	Christoffel's symbols	
Cylinder	$\Gamma_{12}^1 = 0$	$\Gamma_{22}^1 = 0$
Almond	$\Gamma_{22}^1 = 0$	$\Gamma_{22}^1 = \frac{0.5(3 - \bar{v}^2) \sin(2\bar{v}) + \bar{v}(3 \cos^2(\bar{v}) - 1)}{\sin^2(\bar{v}) + (\sin(\bar{v}) + \bar{v} \cos(\bar{v}))^2}$

8. Experimental results

8.1. Hardware description

The proposed modeling framework¹ was applied to the BMWrapArm hardware [9], as depicted in Fig. 12. The system was controlled by two computers: one equipped with an Intel Core i9-11900K CPU @ 3.50 GHz and 32.0 GB of RAM, running MATLAB with the CASPR software for the proposed CWGIK and CWGFK framework; the other, a 64-bit system with a 3.4-GHz Core i7-6700 processor and 16-GB RAM, running the CASPR-ROS software platform for low-level motor control [32].

The ROS network, operating at 20 Hz, managed the transmission of cable lengths to and from the BMWrapArm. Motor encoders on the BMWrapArm were used to obtain cable length feedback $\bar{l}$. The BMWrapArm's base and mobile-platform feature a modular interface, enabling the attachment of various types of mobile-platforms and obstacle surfaces to validate the modeling framework discussed in Sections 3 and 5. For experimental validation, cone-frustum (Fig. 12(a)) and almond-shaped (Fig. 12(e)) mobile-platforms, as well as a cylindrical-shaped obstacle, were selected to validate the CWGIK model. The obstacle was positioned at ${}^O r_{OS} = [0.0707, 0, 0.0707]^T$ m, which is the origin of the frame $\{F_S\}$. The non-zero Christoffel symbols of the cylinder and almond surfaces are listed in Table 2. All experiments were performed with a sampling time of $T_s = 0.05$ s.

8.2. CWGFK

In order to estimate the BMWrapArm's k th time-step joint position $\bar{q}_k$ from the measured cable length feedback $\bar{l}_k$ in the presence of cable wrapping, a numerical CWGFK model was built. This approach was necessary because the nonlinear relationships between the cables and the mobile-platforms prevent an analytical solution to the CWGFK problem. Hence, $\bar{q}_k$ was obtained by solving the following optimization problem

$$\begin{aligned} \bar{q}_k = \underset{q_k}{\operatorname{argmin}} \quad & \|\bar{l}_k - l_k\|_2^2 \\ \text{subject to} \quad & q_{l,k} \leq q_k \leq q_{u,k}, \end{aligned} \quad (68)$$

¹ A video of the simulations and hardware experiments with the BMWrapArm CDPR is available with this article.

where $\bar{l}_k$ and l_k represent the measured cable length feedback and predicted cable length by the CWGIK model, respectively, at the k th time-step. Solving (68) can obtain the least-square estimation of q_k , denoted as $\tilde{q}_k$. However, in real-time, due to the non-convexity of the problem, solving (68) for every time-step is computationally expensive, hence the solution of (68) was obtained as

$$q_k^{\text{est}} = \begin{cases} \tilde{q}_k, & \text{if } N\%k = 0, \\ q_{k-1}^{\text{est}} + J_k^\dagger \dot{\bar{l}}_k & \text{elsewhere,} \end{cases} \quad (69)$$

where N is the number of time-steps, $\%$ refers to modulo operation, $\dot{\bar{l}}_k$ is the k th time-derivative of the cable length feedback, and $J_k^\dagger$ is the pseudoinverse of J at the k th time-step. In (69), $\tilde{q}_k$ is obtained after every $N = 40$ iterations, and the second case of (69) is used to estimate the CDPR pose at every time-step, when $N\%k \neq 0$.

8.3. CWGIK and CWGFK experiments

This section presents the experimental results of actuating the BMWrapArm using cable commands l , generated through the CWGIK, and the estimated joint-space positions q^{est} , obtained from the CWGFK from length feedback $\bar{l}$. The simulations performed for the experiments assumed a frictionless surface, with the cable moving freely on top of the mobile-platform surface.

Cone-frustum-shaped mobile-platform: For actuating the BMWrapArm hardware with a cone-frustum-shaped mobile-platform, firstly, by using the corresponding CWGIK, length command l shown by the solid lines in Fig. 12(b), was generated offline from a reference joint-space trajectory q shown by the solid lines in Fig. 12(b). Secondly, the cable length command l was applied to the hardware, and length feedback $\bar{l}$ from the motor encoders was recorded (Fig. 12(b) dashed lines). Finally, by inputting the length feedback $\bar{l}$ into the CWGFK, the estimated joint-space position q^{est} (Fig. 12(c)) was obtained, and for comparison, Fig. 12(d) showing the error between q^{est} and q was plotted. The Root Mean Square Error (RMSE) between q and q^{est} was found to be $[0.0192, 0.0367, 0.0120]^T$ rad with a maximum error of $[0.0436, 0.0166, 0.0215]^T$ rad (Fig. 12(d)).

Almond-shaped mobile-platform: Similarly, for the BMWrapArm hardware with an almond-shaped mobile-platform, the length command l shown by the solid lines in Fig. 12(f), was generated offline from the reference joint-space trajectory q shown by the solid lines in Fig. 12(g). The RMSE was found to be $[0.004045, 0.005812, 0.004235]^T$ rad with a maximum error of $[0.00205, 0.00381, 0.00414]^T$ rad (Fig. 12(h)). From 21 s in the simulation video (Video 1, Supplementary Materials), while Cable 1 was expected to slide along the almond-shaped mobile-platform during the hardware experiments to minimize its length, this behavior was not observed in experiments due to significant friction between the cable and the mobile-platform surface. The high surface curvature at the top further increased friction, which was reduced to some extent by adding the paper tape.

In summary, the CWGIK model effectively actuated the BMWrapArm by obtaining precise cable lengths, while the CWGFK successfully estimated the reference joint-space trajectory using length feedback from the motor encoders.

9. Conclusion and future work

9.1. Conclusion

Traditional kinematic and dynamic models for CDPRs avoid cable-object interferences to prevent adverse effects on end-effector trajectory. Allowing cable wrapping can expand the workspace and minimize the physical footprint. Hence, this paper introduces a novel numerical-based kinematic and dynamic cable wrapping modeling framework that proficiently detects and manages cable wrapping around mobile-platform and multiple and obstacles, provided the cables remain taut and maintain consistent contact from the point of entry to the point of exit on the surfaces. The primary advantage of the modeling is to make the CDPR design compact for navigating into confined spaces.

The framework consists of a CWG model that numerically solves a set of geodesic differential equations associated with a known CDPR complex-shaped mobile-platform or obstacle surface. The framework also implements a novel cable object interference detection algorithm, which can detect interference with multiple complex-shaped obstacles and mobile-platform. The CWG model and the interference detection algorithm were used to build the CWGIK and CWGID models.

The modeling framework is first validated through simulations on a 3-DoF BMWrapArm with a cone-frustum-shaped mobile-platform and a teapot-shaped obstacle with various reference joint-space trajectories. Finally, the proposed framework was validated experimentally on the BMWrapArm hardware.

9.2. Limitations

While the proposed method demonstrates strong performance for complex surfaces with sharp corners, such as those generated by NURBS and satisfying twice continuous differentiability, some challenges remain for further investigation. One significant limitation is the method's inability to handle surfaces composed of multiple patches or sharp corners, particularly where discontinuities occur at the boundaries. However, in practice, physical objects like boxes do not have perfectly sharp edges due to manufacturing constraints or material properties; edges are always slightly rounded at some scale. As a result, this method is naturally well-suited to accurately model such realistic object shapes.

9.3. Future work

Future work will focus on developing an enhanced framework for determining CWG on multiple patch surfaces treated as a unified surface, even with discontinuities. This will involve methods to handle surface boundaries seamlessly and ensure consistent behavior across patches. Further studies will consider scenarios involving movable or deformable objects to enhance the framework's robustness and applicability. To enhance the framework's practical applications, the effects of surface properties, such as abrasion caused by cable wrapping, will be investigated. Additionally, the model will account for cable elongation effects, analyzing how elasticity and deformation influence wrapping states and the pose of the end-effector. Potentially, multiple cables may interfere when gliding along a surface, adding complexity to cable routing. This will be addressed in future work through advanced interference detection algorithms. Furthermore, incorporating error quantification into the framework will enable the evaluation of how kinematic errors, cable deformations, and displacement inaccuracies affect the proposed CWG model and interference detection algorithm [33]. The framework will incorporate sampling-based methods to identify workspace barriers [34]. Inspired by the IK-based approach, dense sampling and clustering will be applied to detect critical regions where cable routing faces challenges, such as slack, collisions, or constraints. Finally, future work will focus on improving the computational efficiency of the framework and extending its applicability.

CRedit authorship contribution statement

Dipankar Bhattacharya: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Conceptualization. **T.K. Cheung:** Conceptualization. **Yaxiang Wang:** Methodology. **Darwin Lau:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The work was supported by the Research Grants Council (General Research Fund Reference No. 14203921) and the Innovation and Technology Commission (Innovation and Technology Support Programme Reference No. ITS/347/19FP).

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.mechmachtheory.2025.106092>.

Data availability

https://github.com/bhattner143/CASPR_private_dips_wrapping.git.

References

- [1] J.S. Albus, R.V. Bostelman, N. Dagalakakis, The NIST robocrane, *J. Robot. Syst.* 10 (5) (1993) 709–724.
- [2] S.-R. Oh, K. Mankala, S.K. Agrawal, J.S. Albus, A dual-stage planar cable robot: Dynamic modelling and design of a robust controller with positive inputs, *J. Mech. Des.* 127 (4) (2005) 612–620.
- [3] P. Bosscher, R.L. Williams II, L.S. Bryson, D. Castro-Lacouture, Cable-suspended robotic contour crafting system, *Autom. Constr.* 17 (1) (2007) 45–55.
- [4] Y. Wu, H.H. Cheng, A. Fingrut, K. Crolla, Y. Yam, D. Lau, CU-brick cable-driven robot for automated construction of complex brick structures: From simulation to hardware realisation, in: *Proc. IEEE Int. Conf. Sim. Model. Program. Auton. Robot.*, 2018, pp. 166–173.
- [5] J.-B. Izard, A. Dubor, P.-E. Hervé, E. Cabay, D. Culla, M. Rodriguez, M. Barrado, Large-scale 3D printing with cable-driven parallel robots, *Constr. Robot.* 1 (1) (2017) 69–76.
- [6] Y. Mao, S.K. Agrawal, Design of a cable-driven arm exoskeleton (CAREX) for neural rehabilitation, *IEEE Trans. Robot.* 28 (4) (2012) 922–931.
- [7] I.B. Hamida, M.A. Laribi, A. Mlika, L. Romdhane, S. Zeghloul, G. Carbone, Multi-objective optimal design of a cable driven parallel robot for rehabilitation tasks, *Mech. Mach. Theory* 156 (2021) 104141.
- [8] L.L. Cone, Skycam: An aerial robotic system, *Byte* 10 (10) (1985) 122.
- [9] D. Bhattacharya, Y.P. Chan, S. Shang, Y.S. Chan, Y. Tan, D. Lau, Tri-space operational control of redundant multilink and hybrid cable-driven parallel robots using an iterative-learning-based reactive approach, *IEEE Trans. Control Syst. Technol.* (2023).
- [10] J.-H. Bak, S.W. Hwang, J. Yoon, J.H. Park, J.-O. Park, Collision-free path planning of cable-driven parallel robots in cluttered environments, *Intell. Serv. Robot.* 12 (3) (2019) 243–253.
- [11] M.C. Lei, D. Oetomo, Modelling of cable wrapping phenomenon towards improved cable-driven mechanisms, in: *2013 IEEE/ASME International Conference on Advanced Intelligent Mechatronics, IEEE*, 2013, pp. 649–655.
- [12] M.C. Lei, Cable Wrapping Phenomenon in Cable-Driven Parallel Manipulator (Ph.D. thesis), Department of Mechanical Engineering, The University of Melbourne, University of Melbourne, Department of Mechanical Engineering.
- [13] C. Sun, H. Gao, Z. Liu, S. Xiang, H. Yu, N. Li, Z. Deng, Design and optimization of three-degree-of-freedom planar adaptive cable-driven parallel robots using the cable wrapping phenomenon, *Mech. Mach. Theory* 166 (2021) 104475.
- [14] J.-P. Merlet, D. Daney, Legs interference checking of parallel robots over a given workspace or trajectory, in: *Proceedings 2006 IEEE International Conference on Robotics and Automation, 2006. ICRA 2006.*, IEEE, 2006, pp. 757–762.

- [15] L. Blanchet, J.-P. Merlet, Interference detection for cable-driven parallel robots (CDPRs), in: 2014 IEEE/ASME International Conference on Advanced Intelligent Mechatronics, IEEE, 2014, pp. 1413–1418.
- [16] S. Perreault, P. Cardou, C.M. Gosselin, M.J.-D. Otis, Geometric determination of the interference-free constant-orientation workspace of parallel cable-driven mechanisms, 2010.
- [17] M. Rushton, A. Khajepour, Planar variable structure cable-driven parallel robots for circumventing obstacles, *J. Mech. Robot.* 13 (2) (2021) 021011.
- [18] P. Chesler, Numerical solutions for geodesics on two dimensional surfaces, 1999.
- [19] D. Eberly, Computing geodesics on a Riemannian manifold, 2005.
- [20] M.C. Lei, Dynamics of cable driven parallel manipulator allowing cable wrapping over rigid link, in: 2020 IEEE/ASME International Conference on Advanced Intelligent Mechatronics, AIM, IEEE, 2020, pp. 215–221.
- [21] M.C. Lei, D. Oetomo, Cable wrapping phenomenon in cable-driven parallel manipulators., *J. Robot. Mechatronics* 28 (3) (2016) 386–396.
- [22] H. Xiong, Y. Xu, Statics and path of the cables of a cable-driven parallel robot wrapping on surfaces, in: *International Conference on Cable-Driven Parallel Robots*, Springer, 2023, pp. 82–94.
- [23] H. Xiong, Y. Xu, Y. Yu, Y. Lou, Cable path analysis and kinematic control of a cable-driven parallel robot allowing cables to wrap on cylinders or spheres, *J. Mech. Robot.* (2024) 1–15.
- [24] Y. Zou, Y. Hu, H. Cao, Y. Xu, Y. Yu, W. Lu, H. Xiong, Data-driven kinematic control scheme for cable-driven parallel robots allowing collisions, in: 2022 IEEE/RSJ International Conference on Intelligent Robots and Systems, IROS, IEEE, 2022, pp. 5003–5008.
- [25] J.E. Lloyd, F. Roewer-Després, I. Stavness, Muscle path wrapping on arbitrary surfaces, *IEEE Trans. Biomed. Eng.* 68 (2) (2020) 628–638.
- [26] A. Scholz, M. Sherman, I. Stavness, S. Delp, A. Kecskeméthy, A fast multi-obstacle muscle wrapping method using natural geodesic variations, *Multibody Syst. Dyn.* 36 (2016) 195–219.
- [27] I. Muñoz-Pepi, N. Garcia-Hernandez, V. Parra-Vega, Modeling of muscle wrapping phenomenon: A geometric method based on tangency constraints, *IEEE Robot. Autom. Lett.* (2023).
- [28] D.T.M. Lau, Modelling and Analysis of Anthropomorphic Cable-Driven Robots (Ph.D. thesis), University of Melbourne, Department of Mechanical Engineering, 2014.
- [29] L. Piegl, W. Tiller, *The NURBS Book*, Springer Science & Business Media, 2012.
- [30] D.H. Eberly, Representing a circle or a sphere with NURBS, *Geom. Tools* (2016) URL <https://www.geometrictools.com/Documentation/NURBSCircleSphere.pdf> (Accessed 07 April 2025).
- [31] D.T.M. Lau, Modelling and Analysis of Anthropomorphic Cable-Driven Robots (Ph.D. thesis), The University of Melbourne, Melbourne, Australia, 2014, Department of Mechanical Engineering..
- [32] D. Lau, J. Eden, Y. Tan, D. Oetomo, CASPR: A comprehensive cable-robot analysis and simulation platform for the research of cable-driven parallel robots, in: *Proc. IEEE/RSJ Int. Conf. Intell. Robot. Syst.*, 2016, pp. 3004–3011.
- [33] B. Zhou, S. Wu, B. Zi, W. Zhu, Design and optimization of a cable-driven parallel polishing robot with kinematic error modeling, *J. Mech. Des.* 147 (1) (2025).
- [34] A. Peidro, O. Reinoso, A. Gil, J.M. Marín, L. Paya, A method based on the vanishing of self-motion manifolds to determine the collision-free workspace of redundant robots, *Mech. Mach. Theory* 128 (2018) 84–109.